MA'ADEN معادن

Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**

Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **DESIGN CALCULATION**
3310 ROUGHER AND SCAVENGER FLOTATION
DESIGN OF FOUNDATION

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MD-216-3310-EG-CV-CAL-1194	**Mansourah Massarah Gold Project**	
900707_GAU01_ZF01_CED01_33102 _OU100665189_R2	**DESIGN CALCULATION ROUGHER AND SCAVENGERFLOTATION DESIGN OF FOUNDATION**	**Date: 09.11.2023**
O19045_450_C_K_ 24085 _001_R2	**AS BUILT**	**Rev: D01**

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DEFINITIONS AND ABBREVIATIONS

Employer	MA'ADEN GOLD AND BASE METALS or their representatives
EPC Contractor	Consortium of Outotec (EPC-OT) and Larsen & Toubro (EPC-L&T)
Project management Company (PMC)	BECHTEL appointed by EMPLOYER
Supplier	An entity (supplier, consultant, contractor) under the direct project management of EPC Contractor
Sub- supplier	A sub entity (supplier, consultant, contractor) under the direct project management of EPC Contractor
Shall	Is to be understood as mandatory in relation to the requirements of this Document
Should	Is to be understood as a strong recommendation to comply with the requirements of this Document

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1.0 INTENT

This document is to prepare design of **Rougher and Scavenger Flotation (3310)** Structure Foundation. The analysis & design is carried out in accordance with project specifications and codes & standards. The design calculation ensures the safety and serviceability of the foundation system.

2.0 SCOPE

The analysis has been carried out in Excel. For member strength calculation force values are directly taken from STAAD Pro & designed in excel sheet.

3.0 DESIGN REQUIREMENTS

3.1 Design Load

All Design Loads are obtained from STAAD analysis of structure (Doc. No. MD-216-3310-EG-ST-CAL- 2177) attached in Appendix A. Load on structure has been considered as per Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.2 and OTE supplied loads of cells.

4.0 CONCRETE FOUNDATIONS

4.1 Design

Design of concrete structures and foundations shall be in accordance with SBC-304 and ACI 318M.

4.2 Materials

Concrete Strength of 28 days

- Foundation and Structural Members: 35 MPa
- Grade slab and pavement: 30 MPa
- Blinding Concrete below foundation bases: 15 Mpa

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Reinforcement Grade

- Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
- Welded Wire Fabric : ASTM A 185(Plain) / A 497(Deformed)

4.3 Foundations for Structures

Foundation shall be Isolated Shallow Concrete foundation. The design calculations are prepared as per recommendation of standard SBC-304 Clause 13.3.

4.4 Minimum Concrete Cover

Concrete Exposure	Member	Minimum Cover(mm)
Exposed to weather or contact with ground	Foundation & Columns	50

4.5 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundations design has been considered as 450 kN/m^2 at 1.5m depth below EGL (ie. RL 911.50m) refer Table 5 - BH-10 of Geotechnical Report (MD-216-3000-EG-CV-RPT-1008).

4.6 Reference

Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.1.

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DESIGN CALCULATION FOR FOUNDATION

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DESIGN OF FOUNDATION MKD. F1 (STAAD Node -7006)

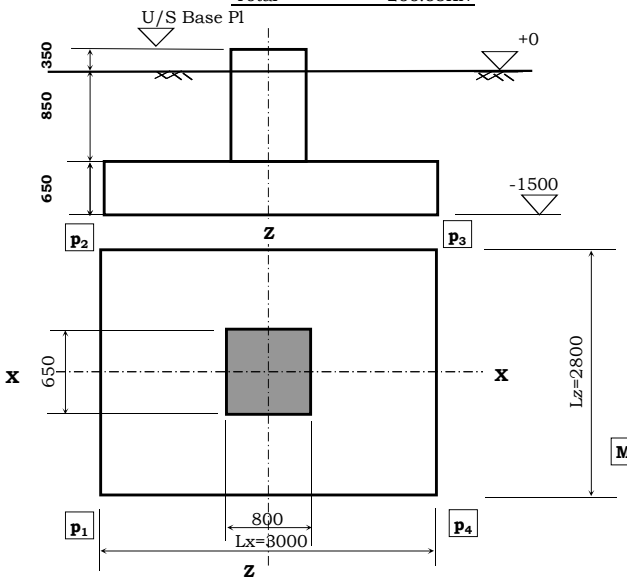
	Along-X	Along-Z
Size of column :	0.80m	0.65m
Size of fdn. raft :	3.00m	2.80m
Thickness of raft :	0.65 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation	-1.50 m	
U/S level Base Plate :	0.35 m	
Unit weight of conc :	24 kN/m^3	
Unit weight of soil :	18 kN/m^3	

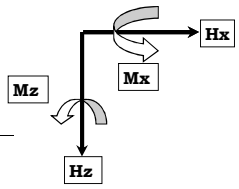
SYMBOLS AND NOTATIONS

Dead Load	DL
Normal Operational Fluid Load	FL1
Max. Fluid Load	FL2
Wind Load	WL
Seismic Load	SL

Calculation of Weight of Foundation :

Weight of raft & Col.=	(3*2.8*0.65*24)+(0.8*0.65*1.2*24) =	146.02KN
Weight of soil =	((3*2.8)-(0.8*0.65))*0.85*18 =	120.56KN
	Total =	266.58KN





Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	20mm
Dia. of bars provd. along X direction at top =	
Dia. of bars provided along Z direction at bottom=	20mm
Dia. of bars provided along Z direction at top=	
Mininum percentage of reinforcement at bottom=	0.18%

Effective depth of slab along X direction = 590mm = 0.59m

Effective depth of slab along Z direction = 570mm = 0.57m

Clear projection along Z direction = 1.075m

Clear projection along X direction = 1.1m

H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

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DESIGN OF FOUNDATION MKD. F1

Combined Load Cases:		Load at top of foundation					Load at bottom of foundation				
Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Stress Design											
201:1.0DL+1.0FL1		2002.8	99.4	0.0	-10.0	0.0	2269.41	99.37	183.83	-10.00	-18.50
202:1.0DL+1.0FL2		2359.1	121.3	0.0	-11.2	0.0	2625.72	121.31	224.42	-11.22	-20.76
203:1.0DL+1.0FL1+1.0LL+1.0TL		2095.7	115.1	0.0	19.1	0.0	2362.27	115.05	212.84	19.07	35.28
204:1.0DL+1.0FL2+1.0LL+1.0TL		2452.0	137.0	0.0	17.9	0.0	2718.58	137.00	253.45	17.85	33.02
205:1.0DL+1.0FL1+0.75LL+0.75TL		2072.5	111.1	0.0	11.8	0.0	2339.05	111.13	205.59	11.80	21.83
206:1.0DL+1.0FL2+0.75LL+0.75TL		2428.8	133.1	0.0	10.6	0.0	2695.36	133.08	246.20	10.58	19.57
207:1.0DL+1.0FL1+1.0PL		1927.6	117.3	0.0	3.3	0.0	2194.16	117.34	217.08	3.31	6.12
208:1.0DL+1.0FL2+1.0PL		2283.9	139.3	0.0	2.1	0.0	2550.47	139.29	257.69	2.08	3.85
209:1.0DL+1.0FL1+1.0LL+1.0TL+1.0PL		2020.4	133.0	0.0	32.4	0.0	2287.02	133.03	246.11	32.38	59.90
210:1.0DL+1.0FL2+1.0LL+1.0TL+1.0PL		2376.8	155.0	0.0	31.2	0.0	2643.33	154.98	286.71	31.15	57.63
211:1.0DL+1.0FL1+0.75LL+0.75TL+1.0PL		1997.2	129.1	0.0	25.1	0.0	2263.80	129.11	238.85	25.11	46.45
212:1.0DL+1.0FL2+0.75LL+0.75TL+1.0PL		2353.5	151.1	0.0	23.9	0.0	2620.11	151.06	279.46	23.89	44.20
213:1.0DL+1.0FL1+1.0WLX1		2012.9	60.7	0.0	-9.1	0.0	2279.50	60.69	112.28	-9.13	-16.89
214:1.0DL+1.0FL1+1.0WLX2		1992.7	138.0	0.0	-10.9	0.0	2259.31	138.04	255.37	-10.87	-20.11
215:1.0DL+1.0FL1+1.0WLZ1		2000.4	93.3	0.0	-25.8	0.0	2266.93	93.30	172.61	-25.81	-47.75
216:1.0DL+1.0FL1+1.0WLZ2		2005.3	105.4	0.0	5.8	0.0	2271.89	105.43	195.05	5.81	10.75
217:1.0DL+1.0FL2+1.0WLX1		2369.2	82.6	0.0	-10.4	0.0	2635.81	82.64	152.88	-10.35	-19.15
218:1.0DL+1.0FL2+1.0WLX2		2349.0	160.0	0.0	-12.1	0.0	2615.62	159.99	295.98	-12.10	-22.39
219:1.0DL+1.0FL2+1.0WLZ1		2356.7	115.3	0.0	-27.0	0.0	2623.24	115.25	213.21	-27.03	-50.01
220:1.0DL+1.0FL2+1.0WLZ2		2361.6	127.4	0.0	4.6	0.0	2628.20	127.38	235.65	4.59	8.49
221:1.0DL+1.0FL1+0.7SLX1		2012.8	8.5	0.0	-9.9	0.0	2279.39	8.49	15.71	-9.90	-18.32
222:1.0DL+1.0FL1+0.7SLX2		1992.8	190.3	0.0	-10.1	0.0	2259.42	190.25	351.96	-10.10	-18.69
223:1.0DL+1.0FL1+0.7SLZ1		2113.7	82.9	0.0	-76.9	0.0	2380.26	82.93	153.42	-76.91	-142.28
224:1.0DL+1.0FL1+0.7SLZ2		1892.0	115.8	0.0	56.9	0.0	2158.56	115.81	214.25	56.91	105.28
225:1.0DL+1.0FL2+0.7SLX1		2369.1	30.4	0.0	-11.1	0.0	2635.70	30.43	56.30	-11.13	-20.59
226:1.0DL+1.0FL2+0.7SLX2		2349.2	212.2	0.0	-11.3	0.0	2615.73	212.19	392.55	-11.32	-20.94
227:1.0DL+1.0FL2+0.7SLZ1		2470.0	104.9	0.0	-78.1	0.0	2736.57	104.88	194.03	-78.13	-144.54
228:1.0DL+1.0FL2+0.7SLZ2		2248.3	137.8	0.0	55.7	0.0	2514.87	137.75	254.84	55.68	103.01
229:1.0DL+1.0FL1+0.75WLX1+0.75LL		2128.4	75.5	0.0	-8.0	0.0	2395.02	75.52	139.71	-8.00	-14.80
230:1.0DL+1.0FL1+0.75WLX2+0.75LL		2113.3	133.5	0.0	-9.3	0.0	2379.88	133.54	247.05	-9.31	-17.22
231:1.0DL+1.0FL1+0.75WLZ1+0.75LL		2119.0	100.0	0.0	-20.5	0.0	2385.59	99.99	184.98	-20.51	-37.94
232:1.0DL+1.0FL1+0.75WLZ2+0.75LL		2122.7	109.1	0.0	3.2	0.0	2389.31	109.08	201.80	3.20	5.92

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233:1.0DL+1.0FL2+0.75WLX1+0.75LL	2484.8	97.5	0.0	-9.2	0.0	2751.33	97.47	180.32	-9.23	-17.08
234:1.0DL+1.0FL2+0.75WLX2+0.75LL	2469.6	155.5	0.0	-10.5	0.0	2736.19	155.49	287.66	-10.54	-19.50
235:1.0DL+1.0FL2+0.75WLZ1+0.75LL	2475.3	121.9	0.0	-21.7	0.0	2741.90	121.93	225.57	-21.74	-40.22
236:1.0DL+1.0FL2+0.75WLZ2+0.75LL	2479.0	131.0	0.0	2.0	0.0	2745.62	131.03	242.41	1.98	3.66
237:1.0DL+1.0FL1+0.525SLX1+0.75LL	2128.4	36.4	0.0	-8.6	0.0	2394.94	36.37	67.28	-8.58	-15.87
238:1.0DL+1.0FL1+0.525SLX2+0.75LL	2113.4	172.7	0.0	-8.7	0.0	2379.96	172.69	319.48	-8.73	-16.15
239:1.0DL+1.0FL1+0.525SLZ1+0.75LL	2204.0	92.2	0.0	-58.8	0.0	2470.59	92.20	170.57	-58.84	-108.85
240:1.0DL+1.0FL1+0.525SLZ2+0.75LL	2037.7	116.9	0.0	41.5	0.0	2304.31	116.86	216.19	41.52	76.81
241:1.0DL+1.0FL2+0.525SLX1+0.75LL	2484.7	58.3	0.0	-9.8	0.0	2751.25	58.32	107.89	-9.81	-18.15
242:1.0DL+1.0FL2+0.525SLX2+0.75LL	2469.7	194.6	0.0	-10.0	0.0	2736.27	194.64	360.08	-9.96	-18.43
243:1.0DL+1.0FL2+0.525SLZ1+0.75LL	2560.3	114.2	0.0	-60.1	0.0	2826.90	114.15	211.18	-60.06	-111.11
244:1.0DL+1.0FL2+0.525SLZ2+0.75LL	2394.0	138.8	0.0	40.3	0.0	2660.62	138.81	256.80	40.30	74.56
245:0.6DL+1.0WLX1	319.9	-21.5	0.0	-2.0	0.0	586.43	-21.51	-39.79	-2.01	-3.72
246:0.6DL+1.0WLX2	299.7	55.9	0.0	-3.8	0.0	566.24	55.85	103.32	-3.75	-6.94
247:0.6DL+1.0WLZ1	307.3	11.1	0.0	-18.7	0.0	573.86	11.11	20.55	-18.69	-34.58
248:0.6DL+1.0WLZ2	312.2	23.2	0.0	12.9	0.0	578.82	23.23	42.98	12.93	23.92
249:0.6DL+0.6FL1+0.7SLX1	1211.7	-31.3	0.0	-5.9	0.0	1478.26	-31.26	-57.83	-5.90	-10.92
250:0.6DL+0.6FL1+0.7SLX2	1191.7	150.5	0.0	-6.1	0.0	1458.29	150.50	278.43	-6.10	-11.29
251:0.6DL+0.6FL1+0.7SLZ1	1312.6	43.2	0.0	-72.9	0.0	1579.13	43.18	79.88	-72.91	-134.88
252:0.6DL+0.6FL1+0.7SLZ2	1090.9	76.1	0.0	60.9	0.0	1357.43	76.06	140.71	60.91	112.68
Ultimate Conditions										
101:1.4DL+1.4FL1	2804.0	139.1	0.0	-14.0	0.0	3070.54	139.11	257.35	-14.00	-25.90
102:1.4DL+1.4FL2	3302.8	169.8	0.0	-15.7	0.0	3569.37	169.84	314.20	-15.71	-29.06
103:1.2DL+1.2FL1+1.6LL	2655.2	130.3	0.0	-9.1	0.0	2921.80	130.26	240.98	-9.14	-16.91
104:1.2DL+1.2FL2+1.6LL	3082.8	156.6	0.0	-10.6	0.0	3349.37	156.60	289.71	-10.61	-19.63
105:1.2DL+1.2FL1+1.2TL+1.6LL	2577.8	140.8	0.0	23.6	0.0	2844.36	140.82	260.52	23.60	43.66
106:1.2DL+1.2FL2+1.2TL+1.6LL	3005.4	167.2	0.0	22.1	0.0	3271.93	167.15	309.23	22.13	40.94
107:1.4DL+1.4FL1+1.4PL	2698.6	164.3	0.0	4.6	0.0	2965.19	164.28	303.92	4.63	8.57
108:1.4DL+1.4FL2+1.4PL	3197.4	195.0	0.0	2.9	0.0	3464.02	195.01	360.77	2.91	5.38
109:1.2DL+1.2FL1+1.6LL+1.2PL	2564.9	151.8	0.0	6.8	0.0	2831.50	151.83	280.89	6.83	12.64
110:1.2DL+1.2FL2+1.6LL+1.2PL	2992.5	178.2	0.0	5.4	0.0	3259.07	178.17	329.61	5.36	9.92
111:1.2DL+1.2FL1+1.2TL+1.6LL+1.2PL	2487.5	162.4	0.0	39.6	0.0	2754.06	162.39	300.42	39.57	73.20

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112:1.2DL+1.2FL2+1.2TL+1.6LL+1.2PL	2915.1	188.7	0.0	38.1	0.0	3181.64	188.73	349.15	38.10	70.49
113:1.2DL+0.8WLX1	627.6	3.4	0.0	-5.1	0.0	894.17	3.39	6.27	-5.07	-9.38
114:1.2DL+0.8WLX2	611.4	65.3	0.0	-6.5	0.0	878.02	65.28	120.77	-6.46	-11.95
115:1.2DL+0.8WLZ1	617.5	29.5	0.0	-18.4	0.0	884.11	29.49	54.56	-18.41	-34.06
116:1.2DL+0.8WLZ2	621.5	39.2	0.0	6.9	0.0	888.08	39.19	72.50	6.89	12.75
117:1.2DL+1.6WLX1+0.5LL	714.4	-24.1	0.0	-3.5	0.0	980.95	-24.11	-44.60	-3.47	-6.42
118:1.2DL+1.6WLX2+0.5LL	682.1	99.7	0.0	-6.3	0.0	948.64	99.66	184.37	-6.26	-11.58
119:1.2DL+1.6WLZ1+0.5LL	694.3	28.1	0.0	-30.2	0.0	960.83	28.08	51.95	-30.16	-55.80
120:1.2DL+1.6WLZ2+0.5LL	702.2	47.5	0.0	20.4	0.0	968.76	47.48	87.84	20.43	37.80
121:1.2DL+1.0SLX1+0.5LL+0.5FL1	1455.8	-56.7	0.0	-7.3	0.0	1722.34	-56.67	-104.84	-7.33	-13.56
122:1.2DL+1.0SLX2+0.5LL+0.5FL1	1427.2	203.0	0.0	-7.6	0.0	1693.81	202.98	375.51	-7.61	-14.08
123:1.2DL+1.0SLZ1+0.5LL+0.5FL1	1599.9	49.7	0.0	-103.1	0.0	1866.44	49.67	91.89	-103.05	-190.64
124:1.2DL+1.0SLZ2+0.5LL+0.5FL1	1283.1	96.6	0.0	88.1	0.0	1549.72	96.64	178.78	88.11	163.00
125:0.9DL+1.6WLX1	480.8	-36.1	0.0	-2.9	0.0	747.37	-36.13	-66.84	-2.93	-5.42
126:0.9DL+1.6WLX2	448.5	87.6	0.0	-5.7	0.0	715.07	87.64	162.13	-5.72	-10.58
127:0.9DL+1.6WLZ1	460.7	16.1	0.0	-29.6	0.0	727.25	16.05	29.69	-29.62	-54.80
128:0.9DL+1.6WLZ2	468.6	35.5	0.0	21.0	0.0	735.19	35.45	65.58	20.97	38.79
129:0.9DL+1.0SLX1	478.9	-104.1	0.0	-4.2	0.0	745.49	-104.08	-192.55	-4.18	-7.73
130:0.9DL+1.0SLX2	450.4	155.6	0.0	-4.5	0.0	716.95	155.58	287.82	-4.46	-8.25
131:0.9DL+1.0SLZ1	623.0	2.3	0.0	-99.9	0.0	889.58	2.27	4.20	-99.90	-184.82
132:0.9DL+1.0SLZ2	306.3	49.2	0.0	91.3	0.0	572.86	49.23	91.08	91.26	168.83

Check for overturning and sliding:

Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201:1.0DL+1.0FL1		2269.4	99.4	183.8	-10.0	-18.5	18.5	184.0	9.1	90.8
202:1.0DL+1.0FL2		2625.7	121.3	224.4	-11.2	-20.8	17.5	189.7	8.7	93.6
203:1.0DL+1.0FL1+1.0LL+1.0TL		2362.3	115.1	212.8	19.1	35.3	16.6	100.4	8.2	49.5
204:1.0DL+1.0FL2+1.0LL+1.0TL		2718.6	137.0	253.5	17.9	33.0	16.1	123.5	7.9	60.9
205:1.0DL+1.0FL1+0.75LL+0.75TL		2339.1	111.1	205.6	11.8	21.8	17.1	160.7	8.4	79.3
206:1.0DL+1.0FL2+0.75LL+0.75TL		2695.4	133.1	246.2	10.6	19.6	16.4	206.6	8.1	101.9
207:1.0DL+1.0FL1+1.0PL		2194.2	117.3	217.1	3.3	6.1	15.2	537.5	7.5	265.2
208:1.0DL+1.0FL2+1.0PL		2550.5	139.3	257.7	2.1	3.8	14.8	994.2	7.3	490.5
209:1.0DL+1.0FL1+1.0LL+1.0TL+1.0PL		2287.0	133.0	246.1	32.4	59.9	13.9	57.3	6.9	28.3
210:1.0DL+1.0FL2+1.0LL+1.0TL+1.0PL		2643.3	155.0	286.7	31.2	57.6	13.8	68.8	6.8	33.9
211:1.0DL+1.0FL1+0.75LL+0.75TL+1.0PL		2263.8	129.1	238.9	25.1	46.5	14.2	73.1	7.0	36.1

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212:1.0DL+1.0FL2+0.75LL+0.75TL+1.0PL	2620.1	151.1	279.5	23.9	44.2	14.1	88.9	6.9	43.9	
213:1.0DL+1.0FL1+1.0WLX1	2279.5	60.7	112.3	-9.1	-16.9	30.5	202.4	15.0	99.9	
214:1.0DL+1.0FL1+1.0WLX2	2259.3	138.0	255.4	-10.9	-20.1	13.3	168.5	6.5	83.1	
215:1.0DL+1.0FL1+1.0WLZ1	2266.9	93.3	172.6	-25.8	-47.7	19.7	71.2	9.7	35.1	
216:1.0DL+1.0FL1+1.0WLZ2	2271.9	105.4	195.0	5.8	10.7	17.5	317.1	8.6	156.4	
217:1.0DL+1.0FL2+1.0WLX1	2635.8	82.6	152.9	-10.4	-19.1	25.9	206.5	12.8	101.9	
218:1.0DL+1.0FL2+1.0WLX2	2615.6	160.0	296.0	-12.1	-22.4	13.3	175.3	6.5	86.5	
219:1.0DL+1.0FL2+1.0WLZ1	2623.2	115.3	213.2	-27.0	-50.0	18.5	78.7	9.1	38.8	
220:1.0DL+1.0FL2+1.0WLZ2	2628.2	127.4	235.7	4.6	8.5	16.7	464.3	8.3	229.0	
221:1.0DL+1.0FL1+0.7SLX1	2279.4	8.5	15.7	-9.9	-18.3	217.7	186.7	107.4	92.1	
222:1.0DL+1.0FL1+0.7SLX2	2259.4	190.3	352.0	-10.1	-18.7	9.6	181.4	4.8	89.5	
223:1.0DL+1.0FL1+0.7SLZ1	2380.3	82.9	153.4	-76.9	-142.3	23.3	25.1	11.5	12.4	
224:1.0DL+1.0FL1+0.7SLZ2	2158.6	115.8	214.2	56.9	105.3	15.1	30.8	7.5	15.2	
225:1.0DL+1.0FL2+0.7SLX1	2635.7	30.4	56.3	-11.1	-20.6	70.2	192.0	34.6	94.7	
226:1.0DL+1.0FL2+0.7SLX2	2615.7	212.2	392.6	-11.3	-20.9	10.0	187.4	4.9	92.4	
227:1.0DL+1.0FL2+0.7SLZ1	2736.6	104.9	194.0	-78.1	-144.5	21.2	28.4	10.4	14.0	
228:1.0DL+1.0FL2+0.7SLZ2	2514.9	137.8	254.8	55.7	103.0	14.8	36.6	7.3	18.1	
229:1.0DL+1.0FL1+0.75WLX1+0.75LL	2395.0	75.5	139.7	-8.0	-14.8	25.7	242.7	12.7	119.8	
230:1.0DL+1.0FL1+0.75WLX2+0.75LL	2379.9	133.5	247.0	-9.3	-17.2	14.4	207.3	7.1	102.3	
231:1.0DL+1.0FL1+0.75WLZ1+0.75LL	2385.6	100.0	185.0	-20.5	-37.9	19.3	94.3	9.5	46.5	
232:1.0DL+1.0FL1+0.75WLZ2+0.75LL	2389.3	109.1	201.8	3.2	5.9	17.8	605.4	8.8	298.7	
233:1.0DL+1.0FL2+0.75WLX1+0.75LL	2751.3	97.5	180.3	-9.2	-17.1	22.9	241.7	11.3	119.2	
234:1.0DL+1.0FL2+0.75WLX2+0.75LL	2736.2	155.5	287.7	-10.5	-19.5	14.3	210.5	7.0	103.8	
235:1.0DL+1.0FL2+0.75WLZ1+0.75LL	2741.9	121.9	225.6	-21.7	-40.2	18.2	102.3	9.0	50.4	
236:1.0DL+1.0FL2+0.75WLZ2+0.75LL	2745.6	131.0	242.4	2.0	3.7	17.0	1124.3	8.4	554.7	
237:1.0DL+1.0FL1+0.525SLX1+0.75LL	2394.9	36.4	67.3	-8.6	-15.9	53.4	226.3	26.3	111.7	
238:1.0DL+1.0FL1+0.525SLX2+0.75LL	2380.0	172.7	319.5	-8.7	-16.2	11.2	221.0	5.5	109.0	
239:1.0DL+1.0FL1+0.525SLZ1+0.75LL	2470.6	92.2	170.6	-58.8	-108.9	21.7	34.0	10.7	16.8	
240:1.0DL+1.0FL1+0.525SLZ2+0.75LL	2304.3	116.9	216.2	41.5	76.8	16.0	45.0	7.9	22.2	
241:1.0DL+1.0FL2+0.525SLX1+0.75LL	2751.3	58.3	107.9	-9.8	-18.1	38.3	227.4	18.9	112.2	
242:1.0DL+1.0FL2+0.525SLX2+0.75LL	2736.3	194.6	360.1	-10.0	-18.4	11.4	222.8	5.6	109.9	
243:1.0DL+1.0FL2+0.525SLZ1+0.75LL	2826.9	114.2	211.2	-60.1	-111.1	20.1	38.2	9.9	18.8	
244:1.0DL+1.0FL2+0.525SLZ2+0.75LL	2660.6	138.8	256.8	40.3	74.6	15.5	53.5	7.7	26.4	
245:0.6DL+1.0WLX1	586.4	-21.5	-39.8	-2.0	-3.7	22.1	236.6	10.9	116.7	
246:0.6DL+1.0WLX2	566.2	55.9	103.3	-3.8	-6.9	8.2	122.4	4.1	60.4	
247:0.6DL+1.0WLZ1	573.9	11.1	20.6	-18.7	-34.6	41.9	24.9	20.7	12.3	
248:0.6DL+1.0WLZ2	578.8	23.2	43.0	12.9	23.9	20.2	36.3	10.0	17.9	
249:0.6DL+0.6FL1+0.7SLX1	1478.3	-31.3	-57.8	-5.9	-10.9	38.3	203.2	18.9	100.2	
250:0.6DL+0.6FL1+0.7SLX2	1458.3	150.5	278.4	-6.1	-11.3	7.9	193.8	3.9	95.6	
251:0.6DL+0.6FL1+0.7SLZ1	1579.1	43.2	79.9	-72.9	-134.9	29.7	17.6	14.6	8.7	
252:0.6DL+0.6FL1+0.7SLZ2	1357.4	76.1	140.7	60.9	112.7	14.5	18.1	7.1	8.9	
					Min.=	7.9	17.6	3.9	8.7	1.5
						OK	OK	OK	OK	

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DESIGN OF FOUNDATION MKD. F1

Load Combination & Calculation of Base pressure

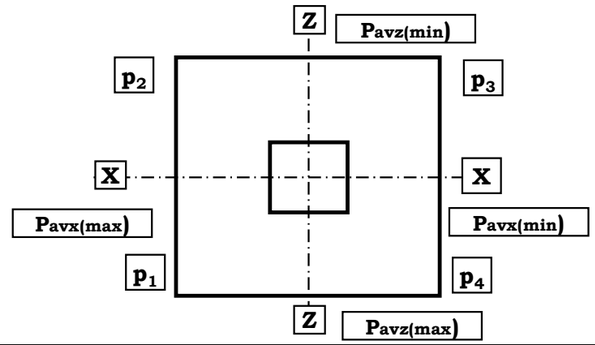
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Surcharge load = 10 kN/m^2

Net SBC of soil for DL, LL= 450 kN/m^2 For WL/SL case = 562.5 kN/m^2

Gross SBC of soil for DL, LL= 477 kN/m^2 For WL/SL case = 589.5 kN/m^2



Allowable Stress Design

Load Case	Load Combination	Design Load/moments V (kN)	Mz (kN-m)	Mx (kN-m)	Base Pressure (kN/m^2) V/ (Lx.Lz)	6.Mz/ (Lx)^2.Lz	6.Mx/ Lx.(Lz)^2	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	$p_{avx(max)}$	$p_{avx(min)}$	$p_{avz(max)}$	$p_{avz(min)}$	Remarks
201:1.0DL+1.0FL1		2269.41	183.83	-18.50	270.17	43.77	4.72	318.66	309.22	231.12	221.68	291.66	313.94	226.40	274.89	265.45	O.K
202:1.0DL+1.0FL2		2625.72	224.42	-20.76	312.59	53.43	5.30	371.32	360.72	264.45	253.86	344.32	366.02	259.16	317.89	307.29	O.K
203:1.0DL+1.0FL1+1.0LL+1.0TL		2362.27	212.84	35.28	281.22	50.68	9.00	340.90	322.90	239.55	221.55	313.90	331.90	230.55	290.23	272.23	O.K
204:1.0DL+1.0FL2+1.0LL+1.0TL		2718.58	253.45	33.02	323.64	60.35	8.42	392.41	375.56	271.72	254.87	365.41	383.99	263.30	332.07	315.22	O.K
205:1.0DL+1.0FL1+0.75LL+0.75TL		2339.05	205.59	21.83	278.46	48.95	5.57	332.98	321.84	235.08	223.94	305.98	327.41	229.51	284.03	272.89	O.K
206:1.0DL+1.0FL2+0.75LL+0.75TL		2695.36	246.20	19.57	320.88	58.62	4.99	384.49	374.50	267.25	257.26	357.49	379.50	262.26	325.87	315.88	O.K
207:1.0DL+1.0FL1+1.0PL		2194.16	217.08	6.12	261.21	51.69	1.56	314.46	311.33	211.09	207.96	287.46	312.90	209.53	262.78	259.65	O.K
208:1.0DL+1.0FL2+1.0PL		2550.47	257.69	3.85	303.63	61.35	0.98	365.96	364.00	243.26	241.29	338.96	364.98	242.28	304.61	302.65	O.K
209:1.0DL+1.0FL1+1.0LL+1.0TL+1.0		2287.02	246.11	59.90	272.26	58.60	15.28	346.14	315.58	228.95	198.39	319.14	330.86	213.67	287.55	256.99	O.K
210:1.0DL+1.0FL2+1.0LL+1.0TL+1.0		2643.33	286.71	57.63	314.68	68.27	14.70	397.65	368.25	261.12	231.72	370.65	382.95	246.42	329.39	299.99	O.K
211:1.0DL+1.0FL1+0.75LL+0.75TL+1		2263.80	238.85	46.45	269.50	56.87	11.85	338.22	314.52	224.48	200.78	311.22	326.37	212.63	281.35	257.65	O.K
212:1.0DL+1.0FL2+0.75LL+0.75TL+1		2620.11	279.46	44.20	311.92	66.54	11.27	389.73	367.18	256.65	234.10	362.73	378.46	245.38	323.19	300.64	O.K
213:1.0DL+1.0FL1+1.0WLX1		2279.50	112.28	-16.89	271.37	26.73	4.31	302.41	293.79	248.95	240.33	275.41	298.10	244.64	275.68	267.06	O.K
214:1.0DL+1.0FL1+1.0WLX2		2259.31	255.37	-20.11	268.97	60.80	5.13	334.90	324.64	213.29	203.03	307.90	329.77	208.16	274.10	263.84	O.K
215:1.0DL+1.0FL1+1.0WLZ1		2266.93	172.61	-47.75	269.87	41.10	12.18	323.15	298.79	240.96	216.60	296.15	310.97	228.78	282.06	257.70	O.K
216:1.0DL+1.0FL1+1.0WLZ2		2271.89	195.05	10.75	270.46	46.44	2.74	319.64	314.16	226.77	221.28	292.64	316.90	224.03	273.21	267.72	O.K
217:1.0DL+1.0FL2+1.0WLX1		2635.81	152.88	-19.15	313.79	36.40	4.88	355.07	345.30	282.27	272.50	328.07	350.19	277.39	318.67	308.90	O.K
218:1.0DL+1.0FL2+1.0WLX2		2615.62	295.98	-22.39	311.38	70.47	5.71	387.57	376.14	246.62	235.20	360.57	381.86	240.91	317.10	305.67	O.K
219:1.0DL+1.0FL2+1.0WLZ1		2623.24	213.21	-50.01	312.29	50.76	12.76	375.81	350.30	274.28	248.77	348.81	363.06	261.53	325.05	299.54	O.K
220:1.0DL+1.0FL2+1.0WLZ2		2628.20	235.65	8.49	312.88	56.11	2.17	371.16	366.82	258.94	254.61	344.16	368.99	256.78	315.05	310.72	O.K

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221:1.0DL+1.0FL1+0.7SLX1	2279.39	15.71	-18.32	271.36	3.74	4.67	279.77	270.42	272.29	262.94	252.77	275.10	267.62	276.03	266.68	O.K
222:1.0DL+1.0FL1+0.7SLX2	2259.42	351.96	-18.69	268.98	83.80	4.77	357.55	348.01	189.94	180.41	330.55	352.78	185.18	273.75	264.21	O.K
223:1.0DL+1.0FL1+0.7SLZ1	2380.26	153.42	-142.28	283.36	36.53	36.30	356.19	283.60	283.13	210.54	329.19	319.90	246.84	319.66	247.07	O.K
224:1.0DL+1.0FL1+0.7SLZ2	2158.56	214.25	105.28	256.97	51.01	26.86	334.84	281.12	232.82	179.10	307.84	307.98	205.96	283.83	230.11	O.K
225:1.0DL+1.0FL2+0.7SLX1	2635.70	56.30	-20.59	313.77	13.40	5.25	332.43	321.92	305.62	295.12	305.43	327.18	300.37	319.03	308.52	O.K
226:1.0DL+1.0FL2+0.7SLX2	2615.73	392.55	-20.94	311.40	93.46	5.34	410.20	399.52	223.27	212.59	383.20	404.86	217.93	316.74	306.06	O.K
227:1.0DL+1.0FL2+0.7SLZ1	2736.57	194.03	-144.54	325.78	46.20	36.87	408.85	335.11	316.46	242.71	381.85	371.98	279.59	362.66	288.91	O.K
228:1.0DL+1.0FL2+0.7SLZ2	2514.87	254.84	103.01	299.39	60.68	26.28	386.34	333.79	264.99	212.44	359.34	360.07	238.72	325.67	273.12	O.K
229:1.0DL+1.0FL1+0.75WLX1+0.75L	2395.02	139.71	-14.80	285.12	33.26	3.78	322.16	314.61	255.63	248.08	295.16	318.39	251.86	288.90	281.35	O.K
230:1.0DL+1.0FL1+0.75WLX2+0.75L	2379.88	247.05	-17.22	283.32	58.82	4.39	346.53	337.75	228.89	220.10	319.53	342.14	224.50	287.71	278.93	O.K
231:1.0DL+1.0FL1+0.75WLZ1+0.75L	2385.59	184.98	-37.94	284.00	44.04	9.68	337.72	318.36	249.64	230.28	310.72	328.04	239.96	293.68	274.32	O.K
232:1.0DL+1.0FL1+0.75WLZ2+0.75L	2389.31	201.80	5.92	284.44	48.05	1.51	334.00	330.98	237.90	234.88	307.00	332.49	236.39	285.95	282.93	O.K
233:1.0DL+1.0FL2+0.75WLX1+0.75L	2751.33	180.32	-17.08	327.54	42.93	4.36	374.83	366.12	288.96	280.25	347.83	370.48	284.61	331.90	323.19	O.K
234:1.0DL+1.0FL2+0.75WLX2+0.75L	2736.19	287.66	-19.50	325.74	68.49	4.97	399.20	389.25	262.22	252.27	372.20	394.23	257.25	330.71	320.76	O.K
235:1.0DL+1.0FL2+0.75WLZ1+0.75L	2741.90	225.57	-40.22	326.42	53.71	10.26	390.38	369.86	282.97	262.45	363.38	380.12	272.71	336.68	316.16	O.K
236:1.0DL+1.0FL2+0.75WLZ2+0.75L	2745.62	242.41	3.66	326.86	57.72	0.93	385.51	383.64	270.08	268.21	358.51	384.58	269.15	327.80	325.93	O.K
237:1.0DL+1.0FL1+0.525SLX1+0.75	2394.94	67.28	-15.87	285.11	16.02	4.05	305.18	297.08	273.14	265.04	278.18	301.13	269.09	289.16	281.06	O.K
238:1.0DL+1.0FL1+0.525SLX2+0.75	2379.96	319.48	-16.15	283.33	76.07	4.12	363.51	355.27	211.38	203.14	336.51	359.39	207.26	287.45	279.21	O.K
239:1.0DL+1.0FL1+0.525SLZ1+0.75	2470.59	170.57	-108.85	294.12	40.61	27.77	362.50	306.96	281.27	225.74	335.50	334.73	253.51	321.89	266.35	O.K
240:1.0DL+1.0FL1+0.525SLZ2+0.75	2304.31	216.19	76.81	274.32	51.47	19.59	345.39	306.20	242.44	203.25	318.39	325.80	222.85	293.92	254.73	O.K
241:1.0DL+1.0FL2+0.525SLX1+0.75	2751.25	107.89	-18.15	327.53	25.69	4.63	357.85	348.59	306.47	297.21	330.85	353.22	301.84	332.16	322.90	O.K
242:1.0DL+1.0FL2+0.525SLX2+0.75	2736.27	360.08	-18.43	325.75	85.73	4.70	416.18	406.78	244.71	235.31	389.18	411.48	240.01	330.45	321.05	O.K
243:1.0DL+1.0FL2+0.525SLZ1+0.75	2826.90	211.18	-111.11	336.54	50.28	28.34	415.16	358.47	314.60	257.91	388.16	386.82	286.26	364.88	308.19	O.K
244:1.0DL+1.0FL2+0.525SLZ2+0.75	2660.62	256.80	74.56	316.74	61.14	19.02	396.90	358.86	274.62	236.58	369.90	377.88	255.60	335.76	297.72	O.K
245:0.6DL+1.0WLX1	586.43	-39.79	-3.72	69.81	9.47	0.95	80.24	78.34	61.29	59.39	53.24	79.29	60.34	70.77	68.87	O.K
246:0.6DL+1.0WLX2	566.24	103.32	-6.94	67.41	24.60	1.77	93.78	90.24	44.58	41.04	66.78	92.01	42.81	69.18	65.64	O.K
247:0.6DL+1.0WLZ1	573.86	20.55	-34.58	68.32	4.89	8.82	82.03	64.39	72.24	54.60	55.03	73.21	63.42	77.14	59.50	O.K
248:0.6DL+1.0WLZ2	578.82	42.98	23.92	68.91	10.23	6.10	85.24	73.04	64.78	52.57	58.24	79.14	58.68	75.01	62.81	O.K
249:0.6DL+0.6FL1+0.7SLX1	1478.26	-57.83	-10.92	175.98	13.77	2.78	192.54	186.97	165.00	159.43	165.54	189.76	162.22	178.77	173.20	O.K
250:0.6DL+0.6FL1+0.7SLX2	1458.29	278.43	-11.29	173.61	66.29	2.88	242.78	237.02	110.19	104.44	215.78	239.90	107.32	176.49	170.73	O.K
251:0.6DL+0.6FL1+0.7SLZ1	1579.13	79.88	-134.88	187.99	19.02	34.41	241.42	172.60	203.38	134.56	214.42	207.01	168.97	222.40	153.58	O.K
252:0.6DL+0.6FL1+0.7SLZ2	1357.43	140.71	112.68	161.60	33.50	28.75	223.85	166.36	156.84	99.35	196.85	195.11	128.10	190.35	132.86	O.K

MA'ADEN معادن		Outotec / LARSEN & TOUBRO
MD-216-3310-EG-CV-CAL-1194	Mansourah Massarah Gold Project	
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O19045_450_C_K_ 24085 _001_R2	AS BUILT	Rev: D01
Ultimate Conditions		

MD-216-3310-EG-CV-CAL-1194				Mansourah Massarah Gold Project													
900707_GAU01_ZF01_CED01_33102 _OU100665189_R2				DESIGN CALCULATION ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION										Date: 09.11.2023			
O19045_450_C_K_ 24085 _001_R2				AS BUILT										Rev: D01			
101:1.4DL+1.4FL1	3070.54	257.35	-25.90	365.54	61.27	6.61	433.42	420.21	310.87	297.66	406.42	426.82	304.27	372.15	358.94		
102:1.4DL+1.4FL2	3569.37	314.20	-29.06	424.93	74.81	7.41	507.15	492.32	357.53	342.70	480.15	499.74	350.12	432.34	417.51		
103:1.2DL+1.2FL1+1.6LL	2921.80	240.98	-16.91	347.83	57.38	4.31	409.52	400.90	294.77	286.14	382.52	405.21	290.46	352.15	343.52		
104:1.2DL+1.2FL2+1.6LL	3349.37	289.71	-19.63	398.73	68.98	5.01	472.72	462.71	334.76	324.75	445.72	467.72	329.76	403.74	393.73		
105:1.2DL+1.2FL1+1.2TL+1.6LL	2844.36	260.52	43.66	338.61	62.03	11.14	411.78	389.50	287.72	265.45	384.78	400.64	276.59	349.75	327.48		
106:1.2DL+1.2FL2+1.2TL+1.6LL	3271.93	309.23	40.94	389.52	73.63	10.44	473.59	452.70	326.33	305.45	446.59	463.15	315.89	399.96	379.08		
107:1.4DL+1.4FL1+1.4PL	2965.19	303.92	8.57	353.00	72.36	2.19	427.55	423.18	282.82	278.45	400.55	425.37	280.64	355.19	350.82		
108:1.4DL+1.4FL2+1.4PL	3464.02	360.77	5.38	412.38	85.90	1.37	499.65	496.91	327.86	325.11	472.65	498.28	326.49	413.76	411.01		
109:1.2DL+1.2FL1+1.6LL+1.2PL	2831.50	280.89	12.64	337.08	66.88	3.22	407.18	400.74	273.43	266.98	380.18	403.96	270.21	340.31	333.86		
110:1.2DL+1.2FL2+1.6LL+1.2PL	3259.07	329.61	9.92	387.98	78.48	2.53	468.99	463.93	312.03	306.98	441.99	466.46	309.51	390.51	385.46		
111:1.2DL+1.2FL1+1.2TL+1.6LL+1.2	2754.06	300.42	73.20	327.86	71.53	18.67	418.07	380.72	275.01	237.66	391.07	399.40	256.34	346.54	309.19		
112:1.2DL+1.2FL2+1.2TL+1.6LL+1.2	3181.64	349.15	70.49	378.77	83.13	17.98	479.88	443.92	313.62	277.65	452.88	461.90	295.64	396.75	360.79		
113:1.2DL+0.8WLX1	894.17	6.27	-9.38	106.45	1.49	2.39	110.33	105.55	107.35	102.56	83.33	107.94	104.96	108.84	104.06		
114:1.2DL+0.8WLX2	878.02	120.77	-11.95	104.53	28.75	3.05	136.33	130.23	78.82	72.72	109.33	133.28	75.77	107.58	101.48		
115:1.2DL+0.8WLZ1	884.11	54.56	-34.06	105.25	12.99	8.69	126.93	109.55	100.95	83.57	99.93	118.24	92.26	113.94	96.56		
116:1.2DL+0.8WLZ2	888.08	72.50	12.75	105.72	17.26	3.25	126.24	119.73	91.71	85.21	99.24	122.99	88.46	108.98	102.47		
117:1.2DL+1.6WLX1+0.5LL	980.95	-44.60	-6.42	116.78	10.62	1.64	129.04	125.76	107.80	104.52	102.04	127.40	106.16	118.42	115.14		
118:1.2DL+1.6WLX2+0.5LL	948.64	184.37	-11.58	112.93	43.90	2.95	159.79	153.88	71.99	66.08	132.79	156.84	69.04	115.89	109.98		
119:1.2DL+1.6WLZ1+0.5LL	960.83	51.95	-55.80	114.38	12.37	14.23	140.99	112.52	116.25	87.78	113.99	126.76	102.02	128.62	100.15		
120:1.2DL+1.6WLZ2+0.5LL	968.76	87.84	37.80	115.33	20.91	9.64	145.88	126.60	104.06	84.77	118.88	136.24	94.42	124.97	105.69		
121:1.2DL+1.0SLX1+0.5LL+0.5FL1	1722.34	-104.84	-13.56	205.04	24.96	3.46	233.46	226.54	183.54	176.62	206.46	230.00	180.08	208.50	201.58		
122:1.2DL+1.0SLX2+0.5LL+0.5FL1	1693.81	375.51	-14.08	201.64	89.41	3.59	294.64	287.46	115.83	108.64	267.64	291.05	112.24	205.24	198.05		
123:1.2DL+1.0SLZ1+0.5LL+0.5FL1	1866.44	91.89	-190.64	222.20	21.88	48.63	292.71	195.44	248.95	151.68	265.71	244.08	200.32	270.83	173.56		
124:1.2DL+1.0SLZ2+0.5LL+0.5FL1	1549.72	178.78	163.00	184.49	42.57	41.58	268.64	185.48	183.51	100.34	241.64	227.06	141.93	226.08	142.91		
125:0.9DL+1.6WLX1	747.37	-66.84	-5.42	88.97	15.91	1.38	106.27	103.50	74.44	71.68	79.27	104.89	73.06	90.36	87.59		
126:0.9DL+1.6WLX2	715.07	162.13	-10.58	85.13	38.60	2.70	126.43	121.03	49.22	43.82	99.43	123.73	46.52	87.83	82.43		
127:0.9DL+1.6WLZ1	727.25	29.69	-54.80	86.58	7.07	13.98	107.63	79.67	93.49	65.53	80.63	93.65	79.51	100.56	72.60		
128:0.9DL+1.6WLZ2	735.19	65.58	38.79	87.52	15.61	9.90	113.03	93.24	81.80	62.01	86.03	103.14	71.91	97.42	77.63		
129:0.9DL+1.0SLX1	745.49	-192.55	-7.73	88.75	45.84	1.97	136.57	132.62	44.88	40.93	109.57	134.60	42.91	90.73	86.78		
130:0.9DL+1.0SLX2	716.95	287.82	-8.25	85.35	68.53	2.10	155.99	151.78	18.93	14.72	128.99	153.89	16.83	87.46	83.25		
131:0.9DL+1.0SLZ1	889.58	4.20	-184.82	105.90	1.00	47.15	154.05	59.76	152.05	57.76	127.05	106.91	104.91	153.05	58.76		
132:0.9DL+1.0SLZ2	572.86	91.08	168.83	68.20	21.68	43.07	132.95	46.81	89.58	3.44	105.95	89.88	46.51	111.27	25.13		

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Max :	3569	393	169	**Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL cases) =**	**291.66 kN/m^2**
Min :	566	-193	-191	**Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL cases) =**	**301.66 kN/m^2**
				Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL+SL/WL cases) =	**389.18 kN/m^2**
				Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL+WL/SL cases) =	**399.18 kN/m^2**

Sample calculation of values of base pressure:

Load case : 202:1.0DL+1.0FL2

Maximum pressure, p_1 (pmax) = 312.59+53.43+5.30= 371.32 kN/m^2

p_2= 312.59+53.43-5.30= 360.72 kN/m^2

p_3= 312.59-(53.43+5.30)= 253.86 kN/m^2

p_4= 312.59-53.43+5.30= 264.45 kN/m^2

Maxm pressure excluding weight of soil (p_{max}-γh)= 371.32-1.5*18= 344.32 kN/m^2

Maximum average Pressure along X, $p_{avx(max)}$ = (371.32+360.72)/2: 366.02 kN/m^2

Minimum average Pressure along X, $p_{avx(minx)}$ = (253.86+264.45)/2: 259.15 kN/m^2

Maximum average Pressure along Y, $p_{avy(max)}$ = (371.32+264.45)/2: 317.88 kN/m^2

Minimum average Pressure along Y, $p_{avy(minx)}$ = (253.86+360.72)/2: 307.29 kN/m^2

DESIGN OF FOUNDATION MKD. F1

Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m^2.

p_{2x} =pressure at the face of column at section 2x-2x in kN/m^2.

p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m^2.

BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.

BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.

SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.

SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((0.8+2*0.5*0.57)+(0.65+2*0.5*0.57))*2

= 5.18 m

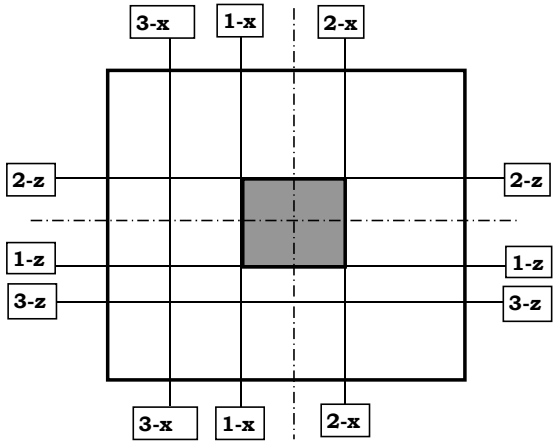
Punching Area= ((0.8+2*0.5*0.57)*(0.65+2*0.5*0.57))

= 1.67 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.65*24+0.85*18)

= 30.90 kN/m^2

Area of footing, A_f= 8.40 m^2



Load Case	Load Combination	p_{1x} kN/m2	p_{2x} kN/m2	p_{3x} kN/m2	B.M_{1x} kNm	B.M_{2x} kNm	S.F_{3x} kN	S.$F_{(punching)}$ kN
Ultimate Conditions								
101:1.4DL+1.4FL1		381.89	349.21	405.99	230.47	174.45	196.61	2252.13
102:1.4DL+1.4FL2		444.88	404.98	474.30	272.58	204.19	232.62	2651.79
103:1.2DL+1.2FL1+1.6LL		363.14	332.54	385.70	217.97	165.52	185.92	2132.96
104:1.2DL+1.2FL2+1.6LL		417.13	380.35	444.27	254.07	191.01	216.80	2475.53
105:1.2DL+1.2FL1+1.2TL+1.6LL		355.16	322.08	379.55	214.52	157.82	183.19	2070.92
106:1.2DL+1.2FL2+1.2TL+1.6LL		409.15	369.89	438.12	250.62	183.31	214.06	2413.48
107:1.4DL+1.4FL1+1.4PL		372.30	333.71	400.77	227.95	161.79	194.91	2167.72
108:1.4DL+1.4FL2+1.4PL		435.29	389.48	469.08	270.06	191.53	230.92	2567.38
109:1.2DL+1.2FL1+1.6LL+1.2PL		354.92	319.25	381.22	215.81	154.67	184.46	2060.61
110:1.2DL+1.2FL2+1.6LL+1.2PL		408.91	367.06	439.78	251.91	180.16	215.33	2403.18
111:1.2DL+1.2FL1+1.2TL+1.6LL+1		346.94	308.80	375.08	212.36	146.97	181.73	1998.57

112:1.2DL+1.2FL2+1.2TL+1.6LL+1	400.94	356.60	433.64	248.46	172.46	212.60	2341.14	

113:1.2DL+0.8WLX1	106.85	106.05	107.43	46.39	45.03	39.16	508.44
114:1.2DL+0.8WLX2	112.19	96.86	123.50	57.69	31.40	49.72	495.50
115:1.2DL+0.8WLZ1	108.71	101.79	113.82	50.92	39.04	43.42	500.38
116:1.2DL+0.8WLZ2	110.33	101.12	117.12	53.16	37.38	45.47	503.56
117:1.2DL+1.6WLX1+0.5LL	119.61	113.95	123.79	56.81	47.10	48.29	577.97
118:1.2DL+1.6WLX2+0.5LL	124.65	101.23	141.91	69.70	29.57	60.42	552.08
119:1.2DL+1.6WLZ1+0.5LL	117.69	111.09	122.55	56.17	44.86	47.82	561.85
120:1.2DL+1.6WLZ2+0.5LL	120.91	109.75	129.13	60.64	41.52	51.91	568.20
121:1.2DL+1.0SLX1+0.5LL+0.5FL1	211.70	198.38	221.51	116.76	93.95	99.38	1171.97
122:1.2DL+1.0SLX2+0.5LL+0.5FL1	225.49	177.80	260.65	144.17	62.43	124.93	1149.11
123:1.2DL+1.0SLZ1+0.5LL+0.5FL1	228.03	216.37	236.64	125.74	105.73	106.82	1287.42
124:1.2DL+1.0SLZ2+0.5LL+0.5FL1	195.85	173.14	212.59	112.38	73.47	96.35	1033.66
125:0.9DL+1.6WLX1	93.22	84.73	99.48	42.41	27.86	36.36	390.83
126:0.9DL+1.6WLX2	95.42	74.83	110.60	50.45	15.16	44.00	364.95
127:0.9DL+1.6WLZ1	88.47	84.69	91.25	36.92	30.45	31.39	374.71
128:0.9DL+1.6WLZ2	91.69	83.36	97.83	41.40	27.12	35.49	381.07
129:0.9DL+1.0SLX1	100.98	76.53	119.01	55.96	14.05	48.91	389.32
130:0.9DL+1.0SLX2	103.63	67.09	130.59	64.27	1.62	56.78	366.46
131:0.9DL+1.0SLZ1	106.18	105.64	106.57	45.84	44.92	38.68	504.77
132:0.9DL+1.0SLZ2	73.98	62.41	82.51	32.48	12.65	28.20	251.01

Sample calculation of BM & SF

Load case : 101:1.4DL+1.4FL1

p_{1x} = 426.82-(426.82-304.27)*1.1/3 = 381.89 kN/m^2 Overhang = 1.1 m

p_{2x} = 304.27+(426.82-304.27)*1.1/3 = 349.21 kN/m^2 Effective depth = 0.59 m

p_{3x} = 426.82-(426.82-304.27)*(1.1-0.59)/3 = 405.99 kN/m^2

BM_{1x} = (381.89-30.9)*1.1*1.1/2+0.5*(426.82-381.89)*1.1*1.1*0.67 = 230.47 kN-m/m

BM_{2x} = (349.21-30.9)*1.1*1.1/2-0.5*(-304.27+349.21)*1.1*1.1*0.67 = 174.45 kN-m/m

SF_{3x} = (405.99-30.9)*(1.1-0.59)+0.5*(426.82-405.99)*(1.1-0.59) = 196.61 kN/m

$S.F_{(punching)}$ = (8.4-1.67)*(365.54-30.9) = 2252.13 kN

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Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	272.58	0.78	1.04	0.003	1493.37	1062.00	20ϕT	210.37	300	200	1570.80

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc)/bd = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 232.62 kN/m

Shear stress , τ_v= (232.62*1000)/(1000*590)

= 0.39 N/mm^2 **O.K.**

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 2651.79 kN

Strength of Concrete in shear Vc1 = 7910.32 kN **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 6445.77 kN **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 6026.91 kN **Table 22.6.5.2_ (c)**

Design Vc= 6026.91 kN

Allowable shear stress, (ϕ Vc)/bd = 1.48 N/mm^2

Design punching shear stress, τ = (2651.79*1000)/(5180*590)

= 0.87 N/mm^2 **No shear reinforcement reqd.**

of Bending Moment and Shear Force in Z direction :

p1z =pressure at the face of column at section 1z-1z in kN/m^2.
p2z =pressure at the face of column at section 2z-2z in kN/m^2.
p3z =pressure at a distance d from the face of column at section 3z-3z in kN/m^2.
B.M1z =Moment at the face of column at section 1z-1z in kN-m/m.
B.M2z =Moment at the face of column at section 2z-2z in kN-m/m.
S.F 3z = Oneway Shear Force at a distance d from the face of column at section 3z-3z in kN/m.
S.F (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Load Case	Load Combination	p_{1z} kN/m2	p_{2z} kN/m2	p_{3z} kN/m2	$B.M_{1z}$ kNm	$B.M_{2z}$ kNm	$S.F_{3z}$ kN
Ultimate Conditions							
101:1.4DL+1.4FL1		367.08	364.01	369.77	196.21	190.51	171.73
102:1.4DL+1.4FL2		426.65	358.10	429.67	230.87	150.95	202.05
103:1.2DL+1.2FL1+1.6LL		348.84	328.58	350.59	184.99	161.97	161.84
104:1.2DL+1.2FL2+1.6LL		399.90	352.96	401.93	214.70	148.94	187.83
105:1.2DL+1.2FL1+1.2TL+1.6LL		341.20	328.21	345.73	182.61	160.87	160.01
106:1.2DL+1.2FL2+1.2TL+1.6LL		391.94	341.56	396.19	211.72	147.02	185.42
107:1.4DL+1.4FL1+1.4PL		353.51	340.21	354.40	187.06	163.40	163.57
108:1.4DL+1.4FL2+1.4PL		412.70	359.05	413.26	221.02	154.00	193.22
109:1.2DL+1.2FL1+1.6LL+1.2PL		337.83	310.92	339.15	178.31	143.57	155.96
110:1.2DL+1.2FL2+1.6LL+1.2PL		388.57	345.76	389.60	207.42	147.84	181.37
111:1.2DL+1.2FL1+1.2TL+1.6LL+1		332.20	295.38	339.80	179.65	142.11	157.70
112:1.2DL+1.2FL2+1.2TL+1.6LL+1		382.94	344.62	390.26	208.76	167.44	183.12

113:1.2DL+0.8WLX1	107.00	185.85	107.98	44.68	135.92	39.14
114:1.2DL+0.8WLX2	105.24	187.31	106.48	43.86	133.82	38.45
115:1.2DL+0.8WLZ1	107.27	180.44	110.81	46.71	136.84	41.14
116:1.2DL+0.8WLZ2	106.48	169.10	107.81	44.64	117.63	39.13
117:1.2DL+1.6WLX1+0.5LL	117.16	176.03	117.83	50.33	118.00	44.05
118:1.2DL+1.6WLX2+0.5LL	113.62	190.48	114.82	48.68	114.12	42.65
119:1.2DL+1.6WLZ1+0.5LL	117.69	170.67	123.49	54.38	103.77	48.05
120:1.2DL+1.6WLZ2+0.5LL	117.57	187.60	121.49	52.94	137.35	46.63
121:1.2DL+1.0SLX1+0.5LL+0.5FL1	205.84	245.79	207.25	102.11	147.50	89.37
122:1.2DL+1.0SLX2+0.5LL+0.5FL1	202.48	261.25	203.94	100.21	143.81	87.71
123:1.2DL+1.0SLZ1+0.5LL+0.5FL1	233.49	231.96	253.29	131.51	132.11	116.73
124:1.2DL+1.0SLZ2+0.5LL+0.5FL1	194.15	198.96	211.08	106.69	129.00	94.78
125:0.9DL+1.6WLX1	89.30	164.42	89.86	34.15	121.48	29.90
126:0.9DL+1.6WLX2	85.76	163.53	86.86	32.50	119.53	28.50
127:0.9DL+1.6WLZ1	89.83	154.51	95.52	38.20	121.14	33.90
128:0.9DL+1.6WLZ2	89.82	175.25	93.85	36.99	140.68	32.69
129:0.9DL+1.0SLX1	89.21	180.43	90.02	34.28	140.73	30.03
130:0.9DL+1.0SLX2	85.84	180.55	86.70	32.37	138.97	28.37
131:0.9DL+1.0SLZ1	116.85	162.05	136.04	63.68	139.22	57.39
132:0.9DL+1.0SLZ2	78.20	126.50	95.73	40.13	115.07	36.66

Sample calculation of BM & SF

Load case : 101:1.4DL+1.4FL1

p_{1z} = 372.15-(372.15-358.94)*1.075/2.8= 367.08 kN/m^2

p_{2z} = 358.94+(372.15-358.94)*1.075/2.8= 364.01 kN/m^2

p_{3z} = 372.15-(372.15-358.94)*(1.075-0.57)/2.8= 369.77 kN/m^2

Overhang = 1.075 m

Effective depth = 0.57 m

BM_{1z} = (367.08-30.9)*1.075*1.075/2+0.5*(372.15-367.08)*1.075*1.075*0.67 = 196.21 kN-m/m

BM_{2z} = (364.01-30.9)*1.075*1.075/2-0.5*(-358.94+364.01)*1.075*1.075*0.67 = 190.51 kN-m/m

SF_{3z} = (369.77-30.9)*(1.075-0.57)+0.5*(372.15-369.77)*(1.075-0.57) = 171.73 kN/m

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	230.87	0.711	0.947	0.0023	1352.85	1026.00	20ϕT	232.22	300	200	1570.8

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc) = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 202.05 kN/m

Shear stress , τ_v= (202.05*1000)/(1000*570)

= 0.35 N/mm^2 **O.K.**

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :

At Bottom **20ϕT** **200** c/c

Along Z Direction :

At Bottom **20ϕT** **200** c/c

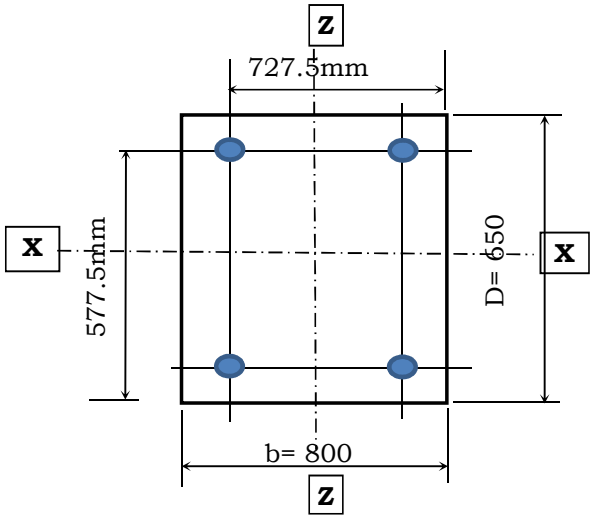
MA'ADEN معادن		Outotec / LARSEN & TOUBRO
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DESIGN OF COLUMN MKD C1 (STAAD Node -7006)

Input:

Clear cover to reinf.=	50mm	Table 20.6.1.3.1
Dia. of main reinf.=	25mm	
Dia. of link=	10mm	
Effective cover =	72.5mm	
Grade of Concrete (f_c') =	M35	
Grade of Steel (f_y) =	Fy420	
Grade of Reinf(links).=	Fy420	
Min. Steel Ratio	0.50%	
Max. Steel Ratio	4.00%	
Self weight of Column =	1.498kN	



For Compression-controlled sections

Strength reduction factor ϕ = 0.65 Table 21.2.2

Unsupproted length of column about X (l_{ux}) =	1200mm
Unsupproted length of column about Z (l_{uz}) =	1200mm

For Columns not braced against sidesway, Where, Effective length factor K= 1.2

Slenderness ratio,	$K\,l_{ux} / r_x$ =	7.68	< 22 SECTION 6.2.5a	Radius of gyration r_x=	188	mm
	$K\,l_{uz} / r_z$ =	6.24	< 22 SECTION 6.2.5a	Radius of gyration r_z=	231	mm

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γ_y =	0.81
γ_z =	0.84

Load Case	Load Combination	Factored Load From STAAD			Load at Column Base			Nominal Load/Moment			Kn = Pn/f_cbD (N/mm^2)
		Pu (T)	Hux (T)	Huz (T)	Pu (T)	Muz (Tm)	Mux (Tm)	Pn (T)	Mnz (Tm)	Mnx (Tm)	
101	1.4DL+1.4FL1	280.40	13.9	-1.4	280.55	16.7	1.7	431.6	25.7	2.6	0.24
102	1.4DL+1.4FL2	330.28	17.0	-1.6	330.43	20.4	1.9	508.4	31.4	2.9	0.28
103	1.2DL+1.2FL1+1.6LL	265.52	13.0	-0.9	265.67	15.6	1.1	408.7	24.0	1.7	0.22
104	1.2DL+1.2FL2+1.6LL	308.28	15.7	-1.1	308.43	18.8	1.3	474.5	28.9	2.0	0.26
105	1.2DL+1.2FL1+1.2TL+1.6LL	257.78	14.1	2.4	257.93	16.9	2.8	396.8	26.0	4.4	0.22
106	1.2DL+1.2FL2+1.2TL+1.6LL	300.54	16.7	2.2	300.68	20.1	2.7	462.6	30.9	4.1	0.25
107	1.4DL+1.4FL1+1.4PL	269.86	16.4	0.5	270.01	19.7	0.6	415.4	30.3	0.9	0.23
108	1.4DL+1.4FL2+1.4PL	319.74	19.5	0.3	319.89	23.4	0.3	492.1	36.0	0.5	0.27
109	1.2DL+1.2FL1+1.6LL+1.2PL	256.49	15.2	0.7	256.64	18.2	0.8	394.8	28.0	1.3	0.22
110	1.2DL+1.2FL2+1.6LL+1.2PL	299.25	17.8	0.5	299.40	21.4	0.6	460.6	32.9	1.0	0.25
111	1.2DL+1.2FL1+1.2TL+1.6LL+1.2PL	248.75	16.2	4.0	248.90	19.5	4.7	382.9	30.0	7.3	0.21
112	1.2DL+1.2FL2+1.2TL+1.6LL+1.2PL	291.51	18.9	3.8	291.66	22.6	4.6	448.7	34.8	7.0	0.25
113	1.2DL+0.8WLX1	62.76	0.3	-0.5	62.91	0.4	0.6	96.8	0.6	0.9	0.05
114	1.2DL+0.8WLX2	61.14	6.5	-0.6	61.29	7.8	0.8	94.3	12.1	1.2	0.05
115	1.2DL+0.8WLZ1	61.75	2.9	-1.8	61.90	3.5	2.2	95.2	5.4	3.4	0.05
116	1.2DL+0.8WLZ2	62.15	3.9	0.7	62.30	4.7	0.8	95.8	7.2	1.3	0.05
117	1.2DL+1.6WLX1+0.5LL	71.44	-2.4	-0.3	71.59	2.9	0.4	110.1	4.5	0.6	0.06
118	1.2DL+1.6WLX2+0.5LL	68.21	10.0	-0.6	68.36	12.0	0.8	105.2	18.4	1.2	0.06
119	1.2DL+1.6WLZ1+0.5LL	69.43	2.8	-3.0	69.57	3.4	3.6	107.0	5.2	5.6	0.06
120	1.2DL+1.6WLZ2+0.5LL	70.22	4.7	2.0	70.37	5.7	2.5	108.3	8.8	3.8	0.06
121	1.2DL+1.0SLX1+0.5LL+0.5FL1	145.58	-5.7	-0.7	145.73	6.8	0.9	224.2	10.5	1.4	0.12

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122:1.2DL+1.0SLX2+0.5LL+0.5FL1	142.72	20.3	-0.8	142.87	24.4	0.9	219.8	37.5	1.4	0.12
123:1.2DL+1.0SLZ1+0.5LL+0.5FL1	159.99	5.0	-10.3	160.14	6.0	12.4	246.4	9.2	19.0	0.14
124:1.2DL+1.0SLZ2+0.5LL+0.5FL1	128.31	9.7	8.8	128.46	11.6	10.6	197.6	17.8	16.3	0.11
125:0.9DL+1.6WLX1	48.08	-3.6	-0.3	48.23	4.3	0.4	74.2	6.7	0.5	0.04
126:0.9DL+1.6WLX2	44.85	8.8	-0.6	45.00	10.5	0.7	69.2	16.2	1.1	0.04
127:0.9DL+1.6WLZ1	46.07	1.6	-3.0	46.22	1.9	3.6	71.1	3.0	5.5	0.04
128:0.9DL+1.6WLZ2	46.86	3.5	2.1	47.01	4.3	2.5	72.3	6.5	3.9	0.04
129:0.9DL+1.0SLX1	47.89	-10.4	-0.4	48.04	12.5	0.5	73.9	19.2	0.8	0.04
130:0.9DL+1.0SLX2	45.04	15.6	-0.4	45.19	18.7	0.5	69.5	28.7	0.8	0.04
131:0.9DL+1.0SLZ1	62.30	0.2	-10.0	62.45	0.3	12.0	96.1	0.4	18.4	0.05
132:0.9DL+1.0SLZ2	30.63	4.9	9.1	30.78	5.9	11.0	47.4	9.1	16.8	0.03

Reinforcement: Provide,

4-Nos	25 mm Dia
4-Nos	25 mm Dia
4-Nos	25 mm Dia

1964.29 Sq mm (At Corner)
1964.29 Sq mm (Along Z axis)
1964.29 Sq mm (Along Y axis)

A_{sc} pvd. = 5892.86 Sq mm

Total A_{sc} = 1.1 %

$\beta = P_n / P_0$ (range is 0.55 to 0.70)

Where, $P_0 = 0.85 f'_c (A_g - A_{sc}) + f_y A_{sc}$ = 1776.969 T SECTION 22.4.2.2

Load Case	From Chart of SP17(M)-R35-420.8		M_{noz}	M_{nox}	β	$\alpha_n = \log(0.5)/\log(\beta)$	Interaction ratio	Remarks
	$R_{nz}=M_{noz}/f'_c b^2 D$	$R_{nx}=M_{nox}/f'_c b^2 D$						
101	0.11	0.11	160.2	130.1	0.55	1.16	0.13	O.K
102	0.12	0.12	174.7	142.0	0.55	1.16	0.15	O.K
103	0.10	0.10	145.6	118.3	0.55	1.16	0.13	O.K
104	0.10	0.10	145.6	118.3	0.55	1.16	0.16	O.K
105	0.10	0.10	145.6	118.3	0.55	1.16	0.16	O.K
106	0.10	0.10	145.6	118.3	0.55	1.16	0.19	O.K
107	0.10	0.10	145.6	118.3	0.55	1.16	0.17	O.K
108	0.10	0.10	145.6	118.3	0.55	1.16	0.20	O.K

109	0.10	0.10	145.6	118.3	0.55	1.16	0.15	O.K
110	0.10	0.10	145.6	118.3	0.55	1.16	0.18	O.K
111	0.10	0.10	145.6	118.3	0.55	1.16	0.20	O.K
112	0.10	0.10	145.6	118.3	0.55	1.16	0.23	O.K
113	0.05	0.05	72.8	59.2	0.55	1.16	0.01	O.K
114	0.05	0.05	72.8	59.2	0.55	1.16	0.14	O.K
115	0.05	0.05	72.8	59.2	0.55	1.16	0.09	O.K
116	0.05	0.05	72.8	59.2	0.55	1.16	0.08	O.K
117	0.05	0.05	72.8	59.2	0.55	1.16	0.04	O.K
118	0.05	0.05	72.8	59.2	0.55	1.16	0.21	O.K
119	0.05	0.05	72.8	59.2	0.55	1.16	0.11	O.K
120	0.05	0.05	72.8	59.2	0.55	1.16	0.13	O.K
121	0.05	0.05	72.8	59.2	0.55	1.16	0.12	O.K
122	0.05	0.05	72.8	59.2	0.55	1.16	0.48	O.K
123	0.05	0.05	72.8	59.2	0.55	1.16	0.36	O.K
124	0.05	0.05	72.8	59.2	0.55	1.16	0.42	O.K
125	0.05	0.05	72.8	59.2	0.55	1.16	0.07	O.K
126	0.05	0.05	72.8	59.2	0.55	1.16	0.18	O.K
127	0.05	0.05	72.8	59.2	0.55	1.16	0.09	O.K
128	0.05	0.05	72.8	59.2	0.55	1.16	0.10	O.K
129	0.05	0.05	72.8	59.2	0.55	1.16	0.22	O.K
130	0.05	0.05	72.8	59.2	0.55	1.16	0.35	O.K
131	0.05	0.05	72.8	59.2	0.55	1.16	0.26	O.K
132	0.05	0.05	72.8	59.2	0.55	1.16	0.32	O.K

Links, 10mm dia @ 300 C/C

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DESIGN OF FOUNDATION MKD. F2 (Staad Node -7093)

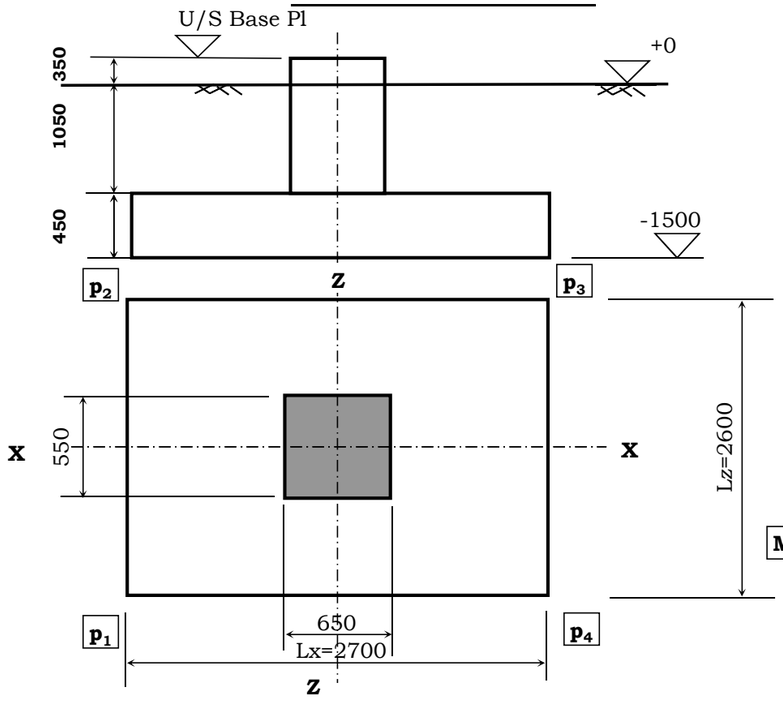
	Along-X	Along-Z
Size of column :	0.65m	0.55m
Size of fdn. raft :	2.70m	2.60m
Thickness of raft :	0.45 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation	-1.50 m	
U/S level Base Plate :	0.35 m	
Unit weight of conc :	24 kN/m^3	
Unit weight of soil :	18 kN/m^3	

SYMBOLS AND NOTATIONS

Dead Load	DL
Normal Operational Fluid Load	FL1
Max. Fluid Load	FL2
Wind Load	WL
Seismic Load	SL

Calculation of Weight of Foundation :

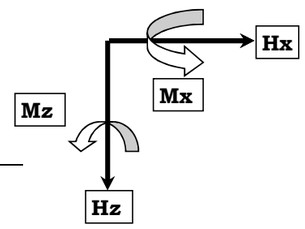
Weight of raft & Col.= (2.7*2.6*0.45*24)+(0.65*0.55*1.4*24) = 87.83KN
Weight of soil = ((2.7*2.6)-(0.65*0.55))*1.05*18 = 125.92KN
Total = 213.75KN



Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f$_y$=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	16mm
Dia. of bars provd. along X direction at top =	
Dia. of bars provided along Z direction at bottom=	16mm
Dia. of bars provided along Z direction at top=	
Mininum percentage of reinforcement at bottom=	0.18%

Effective depth of slab along X direction = 392mm = 0.392m
Effective depth of slab along Z direction = 376mm = 0.376m
Clear projection along Z direction = 1.025m
Clear projection along X direction = 1.025m



H_X=Force along X-X direction
Hz=Force along Z-Z direction
M_X=Moment about X-X direction
Mz=Moment about Z-Z direction

MD-216-3310-EG-CV-CAL-1194	Mansourah Massarah Gold Project	
900707_GAU01_ZF01_CED01_33102 _OU100665189_R2	DESIGN CALCULATION ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION	Date: 09.11.2023
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DESIGN OF FOUNDATION MKD. F2

Combined Load Cases:		Load at top of foundation					Load at bottom of foundation				
Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Stress Design											
201:1.0DL+1.0FL1		338.1	6.0	0.0	-42.5	0.0	551.81	5.97	11.04	-42.54	-78.70
202:1.0DL+1.0FL2		402.3	10.2	0.0	-49.8	0.0	616.02	10.19	18.85	-49.82	-92.17
203:1.0DL+1.0FL1+1.0LL+1.0TL		394.7	-4.8	0.0	-70.1	0.0	608.48	-4.82	-8.92	-70.14	-129.76
204:1.0DL+1.0FL2+1.0LL+1.0TL		458.9	-0.6	0.0	-77.4	0.0	672.69	-0.60	-1.11	-77.42	-143.23
205:1.0DL+1.0FL1+0.75LL+0.75TL		380.6	-2.1	0.0	-63.2	0.0	594.32	-2.13	-3.94	-63.24	-116.99
206:1.0DL+1.0FL2+0.75LL+0.75TL		444.8	2.1	0.0	-70.5	0.0	658.52	2.09	3.87	-70.52	-130.46
207:1.0DL+1.0FL1+1.0PL		404.9	-3.0	0.0	-67.1	0.0	618.61	-3.03	-5.61	-67.09	-124.12
208:1.0DL+1.0FL2+1.0PL		469.1	1.2	0.0	-74.4	0.0	682.81	1.19	2.20	-74.37	-137.58
209:1.0DL+1.0FL1+1.0LL+1.0TL+1.0PL		461.5	-13.8	0.0	-94.7	0.0	675.28	-13.82	-25.57	-94.69	-175.18
210:1.0DL+1.0FL2+1.0LL+1.0TL+1.0PL		525.7	-9.6	0.0	-102.0	0.0	739.48	-9.60	-17.76	-101.97	-188.64
211:1.0DL+1.0FL1+0.75LL+0.75TL+1.0PL		447.4	-11.1	0.0	-87.8	0.0	661.11	-11.12	-20.57	-87.79	-162.41
212:1.0DL+1.0FL2+0.75LL+0.75TL+1.0PL		511.6	-6.9	0.0	-95.1	0.0	725.32	-6.90	-12.77	-95.07	-175.88
213:1.0DL+1.0FL1+1.0WLX1		338.9	-2.2	0.0	-42.9	0.0	552.64	-2.21	-4.09	-42.94	-79.44
214:1.0DL+1.0FL1+1.0WLX2		337.2	14.1	0.0	-42.1	0.0	550.98	14.14	26.16	-42.14	-77.96
215:1.0DL+1.0FL1+1.0WLZ1		486.0	-5.2	0.0	-89.8	0.0	699.71	-5.15	-9.53	-89.84	-166.20
216:1.0DL+1.0FL1+1.0WLZ2		190.2	17.1	0.0	4.8	0.0	403.91	17.08	31.60	4.76	8.81
217:1.0DL+1.0FL2+1.0WLX1		403.1	2.0	0.0	-50.2	0.0	616.85	2.01	3.72	-50.22	-92.91
218:1.0DL+1.0FL2+1.0WLX2		401.4	18.4	0.0	-49.4	0.0	615.19	18.36	33.97	-49.42	-91.43
219:1.0DL+1.0FL2+1.0WLZ1		550.2	-0.9	0.0	-97.1	0.0	763.92	-0.93	-1.72	-97.12	-179.67
220:1.0DL+1.0FL2+1.0WLZ2		254.4	21.3	0.0	-2.5	0.0	468.12	21.30	39.41	-2.52	-4.66
221:1.0DL+1.0FL1+0.7SLX1		334.3	-14.7	0.0	-42.7	0.0	548.00	-14.66	-27.12	-42.65	-78.90
222:1.0DL+1.0FL1+0.7SLX2		341.9	26.6	0.0	-42.4	0.0	555.63	26.59	49.19	-42.43	-78.50
223:1.0DL+1.0FL1+0.7SLZ1		509.9	0.2	0.0	-92.6	0.0	723.68	0.15	0.28	-92.60	-171.31
224:1.0DL+1.0FL1+0.7SLZ2		166.2	11.8	0.0	7.5	0.0	379.95	11.78	21.79	7.53	13.93
225:1.0DL+1.0FL2+0.7SLX1		398.5	-10.4	0.0	-49.9	0.0	612.20	-10.44	-19.31	-49.93	-92.37
226:1.0DL+1.0FL2+0.7SLX2		406.1	30.8	0.0	-49.7	0.0	619.84	30.81	57.00	-49.71	-91.96
227:1.0DL+1.0FL2+0.7SLZ1		574.1	4.4	0.0	-99.9	0.0	787.88	4.37	8.08	-99.88	-184.78
228:1.0DL+1.0FL2+0.7SLZ2		230.4	16.0	0.0	0.3	0.0	444.16	16.00	29.60	0.25	0.46
229:1.0DL+1.0FL1+0.75WLX1+0.75LL		379.7	0.0	0.0	-47.8	0.0	593.45	-0.04	-0.07	-47.78	-88.39
230:1.0DL+1.0FL1+0.75WLX2+0.75LL		378.5	12.2	0.0	-47.2	0.0	592.21	12.22	22.61	-47.18	-87.28
231:1.0DL+1.0FL1+0.75WLZ1+0.75LL		490.0	-2.3	0.0	-83.0	0.0	703.76	-2.25	-4.16	-82.95	-153.46
232:1.0DL+1.0FL1+0.75WLZ2+0.75LL		268.2	14.4	0.0	-12.0	0.0	481.91	14.43	26.70	-12.00	-22.20

233:1.0DL+1.0FL2+0.75WLX1+0.75LL	443.9	4.2	0.0	-55.1	0.0	657.66	4.18	7.73	-55.05	-101.84
234:1.0DL+1.0FL2+0.75WLX2+0.75LL	442.7	16.4	0.0	-54.5	0.0	656.42	16.44	30.41	-54.45	-100.73
235:1.0DL+1.0FL2+0.75WLZ1+0.75LL	554.2	2.0	0.0	-90.2	0.0	767.97	1.97	3.64	-90.23	-166.93
236:1.0DL+1.0FL2+0.75WLZ2+0.75LL	332.4	18.7	0.0	-19.3	0.0	546.12	18.65	34.50	-19.28	-35.67
237:1.0DL+1.0FL1+0.525SLX1+0.75LL	376.2	-9.4	0.0	-47.6	0.0	589.97	-9.38	-17.35	-47.56	-87.99
238:1.0DL+1.0FL1+0.525SLX2+0.75LL	382.0	21.6	0.0	-47.4	0.0	595.70	21.56	39.89	-47.39	-87.67
239:1.0DL+1.0FL1+0.525SLZ1+0.75LL	508.0	1.7	0.0	-85.0	0.0	721.73	1.73	3.20	-85.02	-157.29
240:1.0DL+1.0FL1+0.525SLZ2+0.75LL	250.2	10.5	0.0	-9.9	0.0	463.94	10.45	19.33	-9.93	-18.37
241:1.0DL+1.0FL2+0.525SLX1+0.75LL	440.4	-5.2	0.0	-54.8	0.0	654.18	-5.16	-9.55	-54.84	-101.45
242:1.0DL+1.0FL2+0.525SLX2+0.75LL	446.2	25.8	0.0	-54.7	0.0	659.90	25.78	47.69	-54.67	-101.14
243:1.0DL+1.0FL2+0.525SLZ1+0.75LL	572.2	6.0	0.0	-92.3	0.0	785.94	5.95	11.01	-92.30	-170.76
244:1.0DL+1.0FL2+0.525SLZ2+0.75LL	314.4	14.7	0.0	-17.2	0.0	528.14	14.67	27.14	-17.21	-31.84
245:0.6DL+1.0WLX1	77.4	-6.9	0.0	-10.5	0.0	291.12	-6.93	-12.82	-10.52	-19.46
246:0.6DL+1.0WLX2	75.7	9.4	0.0	-9.7	0.0	289.46	9.42	17.43	-9.72	-17.98
247:0.6DL+1.0WLZ1	224.4	-9.9	0.0	-57.4	0.0	438.19	-9.87	-18.26	-57.42	-106.23
248:0.6DL+1.0WLZ2	-71.4	12.4	0.0	37.2	0.0	142.39	12.36	22.87	37.18	68.78
249:0.6DL+0.6FL1+0.7SLX1	199.0	-17.1	0.0	-25.6	0.0	412.77	-17.05	-31.54	-25.63	-47.42
250:0.6DL+0.6FL1+0.7SLX2	206.7	24.2	0.0	-25.4	0.0	420.40	24.21	44.79	-25.41	-47.01
251:0.6DL+0.6FL1+0.7SLZ1	374.7	-2.2	0.0	-75.6	0.0	588.45	-2.23	-4.13	-75.59	-139.84
252:0.6DL+0.6FL1+0.7SLZ2	31.0	9.4	0.0	24.5	0.0	244.72	9.39	17.37	24.54	45.40
Ultimate Conditions										
101:1.4DL+1.4FL1	473.3	8.4	0.0	-59.6	0.0	687.04	8.35	15.45	-59.55	-110.17
102:1.4DL+1.4FL2	563.2	14.3	0.0	-69.7	0.0	776.93	14.26	26.38	-69.74	-129.02
103:1.2DL+1.2FL1+1.6LL	493.2	7.4	0.0	-61.6	0.0	706.94	7.42	13.73	-61.58	-113.92
104:1.2DL+1.2FL2+1.6LL	570.2	12.5	0.0	-70.3	0.0	783.98	12.49	23.11	-70.31	-130.07
105:1.2DL+1.2FL1+1.2TL+1.6LL	495.6	-5.7	0.0	-86.8	0.0	709.31	-5.72	-10.58	-86.80	-160.58
106:1.2DL+1.2FL2+1.2TL+1.6LL	572.6	-0.7	0.0	-95.5	0.0	786.36	-0.66	-1.22	-95.53	-176.73
107:1.4DL+1.4FL1+1.4PL	566.8	-4.2	0.0	-93.9	0.0	780.55	-4.24	-7.84	-93.93	-173.77
108:1.4DL+1.4FL2+1.4PL	656.7	1.7	0.0	-104.1	0.0	870.44	1.67	3.09	-104.12	-192.62
109:1.2DL+1.2FL1+1.6LL+1.2PL	573.3	-3.4	0.0	-91.1	0.0	787.09	-3.37	-6.23	-91.05	-168.44
110:1.2DL+1.2FL2+1.6LL+1.2PL	650.4	1.7	0.0	-99.8	0.0	864.14	1.70	3.15	-99.78	-184.59
111:1.2DL+1.2FL1+1.2TL+1.6LL+1.2PL	575.7	-16.5	0.0	-116.3	0.0	789.46	-16.51	-30.54	-116.26	-215.08

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112:1.2DL+1.2FL2+1.2TL+1.6LL+1.2PL	652.8	-11.5	0.0	-125.0	0.0	866.51	-11.45	-21.18	-125.00	-231.25
113:1.2DL+0.8WLX1	153.8	-4.0	0.0	-20.6	0.0	367.50	-4.04	-7.47	-20.56	-38.04
114:1.2DL+0.8WLX2	152.4	9.0	0.0	-19.9	0.0	366.17	9.04	16.72	-19.92	-36.85
115:1.2DL+0.8WLZ1	271.4	-6.4	0.0	-58.1	0.0	485.16	-6.40	-11.84	-58.08	-107.45
116:1.2DL+0.8WLZ2	34.8	11.4	0.0	17.6	0.0	248.52	11.39	21.07	17.60	32.56
117:1.2DL+1.6WLX1+0.5LL	181.8	-10.5	0.0	-24.2	0.0	395.51	-10.50	-19.43	-24.18	-44.73
118:1.2DL+1.6WLX2+0.5LL	179.1	15.7	0.0	-22.9	0.0	392.86	15.66	28.97	-22.89	-42.35
119:1.2DL+1.6WLZ1+0.5LL	417.1	-15.2	0.0	-99.2	0.0	630.83	-15.21	-28.14	-99.21	-183.54
120:1.2DL+1.6WLZ2+0.5LL	-56.2	20.4	0.0	52.1	0.0	157.54	20.37	37.68	52.14	96.46
121:1.2DL+1.0SLX1+0.5LL+0.5FL1	280.2	-25.0	0.0	-36.5	0.0	493.98	-24.95	-46.16	-36.53	-67.58
122:1.2DL+1.0SLX2+0.5LL+0.5FL1	291.1	34.0	0.0	-36.2	0.0	504.88	33.99	62.88	-36.21	-66.99
123:1.2DL+1.0SLZ1+0.5LL+0.5FL1	531.2	-3.8	0.0	-107.9	0.0	744.95	-3.78	-6.99	-107.89	-199.60
124:1.2DL+1.0SLZ2+0.5LL+0.5FL1	40.2	12.8	0.0	35.2	0.0	253.91	12.83	23.74	35.15	65.03
125:0.9DL+1.6WLX1	116.1	-11.2	0.0	-15.8	0.0	329.89	-11.21	-20.74	-15.82	-29.27
126:0.9DL+1.6WLX2	113.5	15.0	0.0	-14.5	0.0	327.24	14.95	27.66	-14.54	-26.90
127:0.9DL+1.6WLZ1	351.5	-15.9	0.0	-90.9	0.0	565.21	-15.91	-29.43	-90.86	-168.09
128:0.9DL+1.6WLZ2	-121.8	19.7	0.0	60.5	0.0	91.92	19.66	36.37	60.50	111.93
129:0.9DL+1.0SLX1	109.4	-27.6	0.0	-15.3	0.0	323.11	-27.60	-51.06	-15.34	-28.38
130:0.9DL+1.0SLX2	120.3	31.3	0.0	-15.0	0.0	334.02	31.34	57.98	-15.03	-27.81
131:0.9DL+1.0SLZ1	360.3	-6.4	0.0	-86.7	0.0	574.08	-6.43	-11.90	-86.70	-160.40
132:0.9DL+1.0SLZ2	-130.7	10.2	0.0	56.3	0.0	83.05	10.18	18.83	56.34	104.23

Check for overturning and sliding:

Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201:1.0DL+1.0FL1		551.8	6.0	11.0	-42.5	-78.7	67.4	9.5	37.0	5.2
202:1.0DL+1.0FL2		616.0	10.2	18.9	-49.8	-92.2	44.1	9.0	24.2	4.9
203:1.0DL+1.0FL1+1.0LL+1.0TL		608.5	-4.8	-8.9	-70.1	-129.8	92.1	6.3	50.5	3.5
204:1.0DL+1.0FL2+1.0LL+1.0TL		672.7	-0.6	-1.1	-77.4	-143.2	818.1	6.3	448.5	3.5
205:1.0DL+1.0FL1+0.75LL+0.75TL		594.3	-2.1	-3.9	-63.2	-117.0	203.6	6.9	111.6	3.8
206:1.0DL+1.0FL2+0.75LL+0.75TL		658.5	2.1	3.9	-70.5	-130.5	229.9	6.8	126.0	3.7
207:1.0DL+1.0FL1+1.0PL		618.6	-3.0	-5.6	-67.1	-124.1	149.0	6.7	81.7	3.7
208:1.0DL+1.0FL2+1.0PL		682.8	1.2	2.2	-74.4	-137.6	418.7	6.7	229.5	3.7
209:1.0DL+1.0FL1+1.0LL+1.0TL+1.0PL		675.3	-13.8	-25.6	-94.7	-175.2	35.7	5.2	19.5	2.9
210:1.0DL+1.0FL2+1.0LL+1.0TL+1.0PL		739.5	-9.6	-17.8	-102.0	-188.6	56.2	5.3	30.8	2.9
211:1.0DL+1.0FL1+0.75LL+0.75TL+1.0PL		661.1	-11.1	-20.6	-87.8	-162.4	43.4	5.5	23.8	3.0

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212:1.0DL+1.0FL2+0.75LL+0.75TL+1.0PL	725.3	-6.9	-12.8	-95.1	-175.9	76.7	5.6	42.0	3.1	
213:1.0DL+1.0FL1+1.0WLX1	552.6	-2.2	-4.1	-42.9	-79.4	182.5	9.4	100.0	5.1	
214:1.0DL+1.0FL1+1.0WLX2	551.0	14.1	26.2	-42.1	-78.0	28.4	9.5	15.6	5.2	
215:1.0DL+1.0FL1+1.0WLZ1	699.7	-5.2	-9.5	-89.8	-166.2	99.1	5.7	54.3	3.1	
216:1.0DL+1.0FL1+1.0WLZ2	403.9	17.1	31.6	4.8	8.8	17.3	61.9	9.5	33.9	
217:1.0DL+1.0FL2+1.0WLX1	616.9	2.0	3.7	-50.2	-92.9	223.9	9.0	122.8	4.9	
218:1.0DL+1.0FL2+1.0WLX2	615.2	18.4	34.0	-49.4	-91.4	24.5	9.1	13.4	5.0	
219:1.0DL+1.0FL2+1.0WLZ1	763.9	-0.9	-1.7	-97.1	-179.7	599.4	5.7	328.6	3.1	
220:1.0DL+1.0FL2+1.0WLZ2	468.1	21.3	39.4	-2.5	-4.7	16.0	135.6	8.8	74.3	
221:1.0DL+1.0FL1+0.7SLX1	548.0	-14.7	-27.1	-42.7	-78.9	27.3	9.4	15.0	5.1	
222:1.0DL+1.0FL1+0.7SLX2	555.6	26.6	49.2	-42.4	-78.5	15.2	9.6	8.4	5.2	
223:1.0DL+1.0FL1+0.7SLZ1	723.7	0.2	0.3	-92.6	-171.3	3520.6	5.7	1929.8	3.1	
224:1.0DL+1.0FL1+0.7SLZ2	380.0	11.8	21.8	7.5	13.9	23.5	36.8	12.9	20.2	
225:1.0DL+1.0FL2+0.7SLX1	612.2	-10.4	-19.3	-49.9	-92.4	42.8	8.9	23.5	4.9	
226:1.0DL+1.0FL2+0.7SLX2	619.8	30.8	57.0	-49.7	-92.0	14.7	9.1	8.0	5.0	
227:1.0DL+1.0FL2+0.7SLZ1	787.9	4.4	8.1	-99.9	-184.8	131.6	5.8	72.1	3.2	
228:1.0DL+1.0FL2+0.7SLZ2	444.2	16.0	29.6	0.3	0.5	20.3	1296.5	11.1	710.7	
229:1.0DL+1.0FL1+0.75WLX1+0.75LL	593.5	0.0	-0.1	-47.8	-88.4	10826.5	9.1	5934.5	5.0	
230:1.0DL+1.0FL1+0.75WLX2+0.75LL	592.2	12.2	22.6	-47.2	-87.3	35.4	9.2	19.4	5.0	
231:1.0DL+1.0FL1+0.75WLZ1+0.75LL	703.8	-2.3	-4.2	-83.0	-153.5	228.2	6.2	125.1	3.4	
232:1.0DL+1.0FL1+0.75WLZ2+0.75LL	481.9	14.4	26.7	-12.0	-22.2	24.4	29.3	13.4	16.1	
233:1.0DL+1.0FL2+0.75WLX1+0.75LL	657.7	4.2	7.7	-55.1	-101.8	114.8	8.7	62.9	4.8	
234:1.0DL+1.0FL2+0.75WLX2+0.75LL	656.4	16.4	30.4	-54.5	-100.7	29.1	8.8	16.0	4.8	
235:1.0DL+1.0FL2+0.75WLZ1+0.75LL	768.0	2.0	3.6	-90.2	-166.9	284.5	6.2	155.9	3.4	
236:1.0DL+1.0FL2+0.75WLZ2+0.75LL	546.1	18.7	34.5	-19.3	-35.7	21.4	20.7	11.7	11.3	
237:1.0DL+1.0FL1+0.525SLX1+0.75LL	590.0	-9.4	-17.4	-47.6	-88.0	45.9	9.1	25.2	5.0	
238:1.0DL+1.0FL1+0.525SLX2+0.75LL	595.7	21.6	39.9	-47.4	-87.7	20.2	9.2	11.1	5.0	
239:1.0DL+1.0FL1+0.525SLZ1+0.75LL	721.7	1.7	3.2	-85.0	-157.3	304.4	6.2	166.9	3.4	
240:1.0DL+1.0FL1+0.525SLZ2+0.75LL	463.9	10.5	19.3	-9.9	-18.4	32.4	34.1	17.8	18.7	
241:1.0DL+1.0FL2+0.525SLX1+0.75LL	654.2	-5.2	-9.5	-54.8	-101.5	92.5	8.7	50.7	4.8	
242:1.0DL+1.0FL2+0.525SLX2+0.75LL	659.9	25.8	47.7	-54.7	-101.1	18.7	8.8	10.2	4.8	
243:1.0DL+1.0FL2+0.525SLZ1+0.75LL	785.9	6.0	11.0	-92.3	-170.8	96.4	6.2	52.8	3.4	
244:1.0DL+1.0FL2+0.525SLZ2+0.75LL	528.1	14.7	27.1	-17.2	-31.8	26.3	22.4	14.4	12.3	
245:0.6DL+1.0WLX1	291.1	-6.9	-12.8	-10.5	-19.5	30.7	20.2	16.8	11.1	
246:0.6DL+1.0WLX2	289.5	9.4	17.4	-9.7	-18.0	22.4	21.7	12.3	11.9	
247:0.6DL+1.0WLZ1	438.2	-9.9	-18.3	-57.4	-106.2	32.4	5.6	17.8	3.1	
248:0.6DL+1.0WLZ2	142.4	12.4	22.9	37.2	68.8	8.4	2.8	4.6	1.5	
249:0.6DL+0.6FL1+0.7SLX1	412.8	-17.1	-31.5	-25.6	-47.4	17.7	11.8	9.7	6.4	
250:0.6DL+0.6FL1+0.7SLX2	420.4	24.2	44.8	-25.4	-47.0	12.7	12.1	6.9	6.6	
251:0.6DL+0.6FL1+0.7SLZ1	588.5	-2.2	-4.1	-75.6	-139.8	192.6	5.7	105.6	3.1	
252:0.6DL+0.6FL1+0.7SLZ2	244.7	9.4	17.4	24.5	45.4	19.0	7.3	10.4	4.0	
					Min.=	8.4	2.8	4.6	1.5	1.5
						OK	OK	OK	OK	

MD-216-3310-EG-CV-CAL-1194	Mansourah Massarah Gold Project	
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DESIGN OF FOUNDATION MKD. F2

Load Combination & Calculation of Base pressure

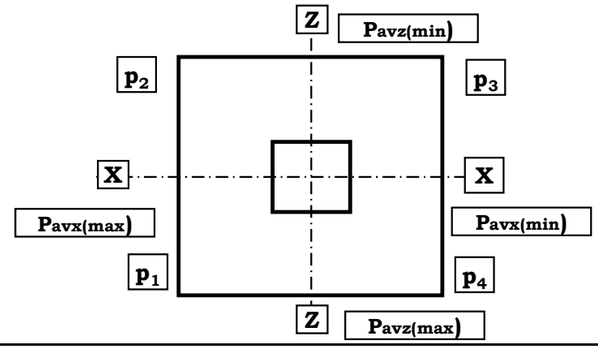
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Surcharge load = 10 kN/m^2

Net SBC of soil for DL, LL= 450 kN/m^2 For WL/SL case = 562.5 kN/m^2

Gross SBC of soil for DL, LL= 477 kN/m^2 For WL/SL case = 589.5 kN/m^2



Allowable Stress Design

Load Case	Load Combination	Design Load/moments			Base Pressure (kN/m^2)												
		V (kN)	Mz (kN-m)	Mx (kN-m)	V/ (Lx.Lz)	6.Mz/ (Lx)^2.Lz	6.Mx/ Lx.(Lz)^2	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	$p_{avx(max)}$	$p_{avx(min)}$	$p_{avz(max)}$	$p_{avz(min)}$	Remarks
201	1.0DL+1.0FL1	551.81	11.04	-78.70	78.61	3.50	25.87	107.97	56.23	100.98	49.24	80.97	82.10	75.11	104.48	52.74	O.K
202	1.0DL+1.0FL2	616.02	18.85	-92.17	87.75	5.97	30.30	124.02	63.42	112.08	51.49	97.02	93.72	81.79	118.05	57.46	O.K
203	1.0DL+1.0FL1+1.0LL+1.0TL	608.48	-8.92	-129.76	86.68	2.82	42.66	132.16	46.84	126.51	41.20	105.16	89.50	83.86	129.34	44.02	O.K
204	1.0DL+1.0FL2+1.0LL+1.0TL	672.69	-1.11	-143.23	95.82	0.35	47.08	143.26	49.09	142.56	48.39	116.26	96.18	95.48	142.91	48.74	O.K
205	1.0DL+1.0FL1+0.75LL+0.75TL	594.32	-3.94	-116.99	84.66	1.25	38.46	124.37	47.45	121.87	44.95	97.37	85.91	83.41	123.12	46.20	O.K
206	1.0DL+1.0FL2+0.75LL+0.75TL	658.52	3.87	-130.46	93.81	1.22	42.89	137.92	52.14	135.47	49.70	110.92	95.03	92.59	136.70	50.92	O.K
207	1.0DL+1.0FL1+1.0PL	618.61	-5.61	-124.12	88.12	1.77	40.80	130.70	49.09	127.15	45.55	103.70	89.90	86.35	128.93	47.32	O.K
208	1.0DL+1.0FL2+1.0PL	682.81	2.20	-137.58	97.27	0.70	45.23	143.19	52.73	141.80	51.34	116.19	97.96	96.57	142.50	52.04	O.K
209	1.0DL+1.0FL1+1.0LL+1.0TL+1.0	675.28	-25.57	-175.18	96.19	8.09	57.59	161.87	46.70	145.69	30.51	134.87	104.29	88.10	153.78	38.61	O.K
210	1.0DL+1.0FL2+1.0LL+1.0TL+1.0	739.48	-17.76	-188.64	105.34	5.62	62.01	172.97	48.95	161.73	37.70	145.97	110.96	99.72	167.35	43.33	O.K
211	1.0DL+1.0FL1+0.75LL+0.75TL+1	661.11	-20.57	-162.41	94.18	6.51	53.39	154.08	47.30	141.05	34.27	127.08	100.69	87.66	147.57	40.79	O.K
212	1.0DL+1.0FL2+0.75LL+0.75TL+1	725.32	-12.77	-175.88	103.32	4.04	57.82	165.18	49.55	157.10	41.46	138.18	107.37	99.28	161.14	45.51	O.K
213	1.0DL+1.0FL1+1.0WLX1	552.64	-4.09	-79.44	78.72	1.29	26.11	106.13	53.90	103.54	51.32	79.13	80.02	77.43	104.84	52.61	O.K
214	1.0DL+1.0FL1+1.0WLX2	550.98	26.16	-77.96	78.49	8.28	25.63	112.40	61.14	95.83	44.58	85.40	86.77	70.21	104.12	52.86	O.K
215	1.0DL+1.0FL1+1.0WLZ1	699.71	-9.53	-166.20	99.67	3.02	54.64	157.33	48.05	151.29	42.02	130.33	102.69	96.66	154.31	45.04	O.K
216	1.0DL+1.0FL1+1.0WLZ2	403.91	31.60	8.81	57.54	10.00	2.89	70.43	64.64	50.43	44.64	43.43	67.54	47.54	60.43	54.64	O.K
217	1.0DL+1.0FL2+1.0WLX1	616.85	3.72	-92.91	87.87	1.18	30.54	119.59	58.51	117.23	56.15	92.59	89.05	86.69	118.41	57.33	O.K
218	1.0DL+1.0FL2+1.0WLX2	615.19	33.97	-91.43	87.63	10.75	30.05	128.44	68.33	106.94	46.83	101.44	98.39	76.89	117.69	57.58	O.K
219	1.0DL+1.0FL2+1.0WLZ1	763.92	-1.72	-179.67	108.82	0.54	59.06	168.43	50.30	167.34	49.21	141.43	109.37	108.28	167.89	49.76	O.K

MD-216-3310-EG-CV-CAL-1194				Mansourah Massarah Gold Project												
900707_GAU01_ZF01_CED01_33102 _OU100665189_R2				DESIGN CALCULATION ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION									Date: 09.11.2023			
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220:1.0DL+1.0FL2+1.0WLZ2	468.12	39.41	-4.66	66.68	12.47	1.53	80.69	77.63	55.74	52.68	53.69	79.16	54.21	68.22	65.16	O.K
221:1.0DL+1.0FL1+0.7SLX1	548.00	-27.12	-78.90	78.06	8.59	25.94	112.59	60.71	95.42	43.54	85.59	86.65	69.48	104.01	52.13	O.K
222:1.0DL+1.0FL1+0.7SLX2	555.63	49.19	-78.50	79.15	15.57	25.80	120.53	68.92	89.38	37.77	93.53	94.73	63.58	104.96	53.35	O.K
223:1.0DL+1.0FL1+0.7SLZ1	723.68	0.28	-171.31	103.09	0.09	56.31	159.49	46.86	159.32	46.69	132.49	103.18	103.01	159.41	46.78	O.K
224:1.0DL+1.0FL1+0.7SLZ2	379.95	21.79	13.93	54.12	6.90	4.58	65.60	56.44	51.80	42.65	38.60	61.02	47.23	58.70	49.55	O.K
225:1.0DL+1.0FL2+0.7SLX1	612.20	-19.31	-92.37	87.21	6.11	30.37	123.69	62.96	111.46	50.73	96.69	93.33	81.10	117.58	56.85	O.K
226:1.0DL+1.0FL2+0.7SLX2	619.84	57.00	-91.96	88.30	18.04	30.23	136.57	76.11	100.48	40.02	109.57	106.34	70.25	118.53	58.07	O.K
227:1.0DL+1.0FL2+0.7SLZ1	787.88	8.08	-184.78	112.23	2.56	60.74	175.54	54.05	170.42	48.93	148.54	114.80	109.68	172.98	51.49	O.K
228:1.0DL+1.0FL2+0.7SLZ2	444.16	29.60	0.46	63.27	9.37	0.15	72.79	72.49	54.05	53.75	45.79	72.64	53.90	63.42	63.12	O.K
229:1.0DL+1.0FL1+0.75WLX1+0.75L	593.45	-0.07	-88.39	84.54	0.02	29.06	113.62	55.50	113.57	55.46	86.62	84.56	84.52	113.60	55.48	O.K
230:1.0DL+1.0FL1+0.75WLX2+0.75L	592.21	22.61	-87.28	84.36	7.16	28.69	120.21	62.82	105.90	48.51	93.21	91.52	77.21	113.06	55.67	O.K
231:1.0DL+1.0FL1+0.75WLZ1+0.75L	703.76	-4.16	-153.46	100.25	1.32	50.45	152.01	51.12	149.38	48.49	125.01	101.57	98.94	150.70	49.81	O.K
232:1.0DL+1.0FL1+0.75WLZ2+0.75L	481.91	26.70	-22.20	68.65	8.45	7.30	84.40	69.80	67.50	52.90	57.40	77.10	60.20	75.95	61.35	O.K
233:1.0DL+1.0FL2+0.75WLX1+0.75L	657.66	7.73	-101.84	93.68	2.45	33.48	129.61	62.65	124.71	57.76	102.61	96.13	91.24	127.16	60.21	O.K
234:1.0DL+1.0FL2+0.75WLX2+0.75L	656.42	30.41	-100.73	93.51	9.63	33.11	136.25	70.02	116.99	50.77	109.25	103.14	83.88	126.62	60.40	O.K
235:1.0DL+1.0FL2+0.75WLZ1+0.75L	767.97	3.64	-166.93	109.40	1.15	54.87	165.42	55.68	163.12	53.37	138.42	110.55	108.25	164.27	54.53	O.K
236:1.0DL+1.0FL2+0.75WLZ2+0.75L	546.12	34.50	-35.67	77.79	10.92	11.73	100.44	76.99	78.60	55.15	73.44	88.72	66.88	89.52	66.07	O.K
237:1.0DL+1.0FL1+0.525SLX1+0.75	589.97	-17.35	-87.99	84.04	5.49	28.92	118.46	60.61	107.47	49.62	91.46	89.54	78.55	112.97	55.12	O.K
238:1.0DL+1.0FL1+0.525SLX2+0.75	595.70	39.89	-87.67	84.86	12.63	28.82	126.30	68.66	101.05	43.41	99.30	97.48	72.23	113.68	56.04	O.K
239:1.0DL+1.0FL1+0.525SLZ1+0.75	721.73	3.20	-157.29	102.81	1.01	51.71	155.53	52.12	153.50	50.09	128.53	103.83	101.80	154.52	51.11	O.K
240:1.0DL+1.0FL1+0.525SLZ2+0.75	463.94	19.33	-18.37	66.09	6.12	6.04	78.25	66.17	66.01	53.93	51.25	72.21	59.97	72.13	60.05	O.K
241:1.0DL+1.0FL2+0.525SLX1+0.75	654.18	-9.55	-101.45	93.19	3.02	33.35	129.56	62.86	123.52	56.82	102.56	96.21	90.17	126.54	59.84	O.K
242:1.0DL+1.0FL2+0.525SLX2+0.75	659.90	47.69	-101.14	94.00	15.10	33.25	142.35	75.85	112.15	45.66	115.35	109.10	78.91	127.25	60.76	O.K
243:1.0DL+1.0FL2+0.525SLZ1+0.75	785.94	11.01	-170.76	111.96	3.48	56.13	171.57	59.31	164.61	52.34	144.57	115.44	108.48	168.09	55.83	O.K
244:1.0DL+1.0FL2+0.525SLZ2+0.75	528.14	27.14	-31.84	75.23	8.59	10.47	94.29	73.36	77.11	56.18	67.29	83.83	66.65	85.70	64.77	O.K
245:0.6DL+1.0WLX1	291.12	-12.82	-19.46	41.47	4.06	6.40	51.93	39.13	43.81	31.01	24.93	45.53	37.41	47.87	35.07	O.K
246:0.6DL+1.0WLX2	289.46	17.43	-17.98	41.23	5.52	5.91	52.66	40.84	41.63	29.81	25.66	46.75	35.72	47.15	35.33	O.K
247:0.6DL+1.0WLZ1	438.19	-18.26	-106.23	62.42	5.78	34.92	103.12	33.28	91.56	21.72	76.12	68.20	56.64	97.34	27.50	O.K
248:0.6DL+1.0WLZ2	142.39	22.87	68.78	20.28	7.24	22.61	50.13	4.91	35.66	0.00	23.13	27.52	17.83	42.90	0.00	O.K
249:0.6DL+0.6FL1+0.7SLX1	412.77	-31.54	-47.42	58.80	9.98	15.59	84.37	53.20	64.40	33.23	57.37	68.79	48.82	74.39	43.22	O.K
250:0.6DL+0.6FL1+0.7SLX2	420.40	44.79	-47.01	59.89	14.18	15.45	89.52	58.61	61.16	30.25	62.52	74.07	45.71	75.34	44.43	O.K
251:0.6DL+0.6FL1+0.7SLZ1	588.45	-4.13	-139.84	83.82	1.31	45.97	131.10	39.16	128.49	36.55	104.10	85.13	82.52	129.80	37.86	O.K

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900707_GAU01_ZF01_CED01_33102 _OU100665189_R2				DESIGN CALCULATION ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION										Date: 09.11.2023			
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252:0.6DL+0.6FL1+0.7SLZ2	244.72	17.37	45.40	34.86	5.50	14.92	55.28	25.44	44.29	14.44	28.28	40.36	29.37	49.79	19.94	O.K	

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900707_GAU01_ZF01_CED01_33102 _OU100665189_R2	DESIGN CALCULATION ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION	Date: 09.11.2023
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Ultimate Conditions

101:1.4DL+1.4FL1	687.04	15.45	-110.17	97.87	4.89	36.22	138.97	66.54	129.19	56.76	111.97	102.76	92.98	134.08	61.65	
102:1.4DL+1.4FL2	776.93	26.38	-129.02	110.67	8.35	42.41	161.44	76.61	144.74	59.91	134.44	119.03	102.33	153.09	68.26	
103:1.2DL+1.2FL1+1.6LL	706.94	13.73	-113.92	100.70	4.35	37.45	142.50	67.60	133.81	58.91	115.50	105.05	96.36	138.16	63.26	
104:1.2DL+1.2FL2+1.6LL	783.98	23.11	-130.07	111.68	7.31	42.76	161.75	76.23	147.12	61.60	134.75	118.99	104.36	154.44	68.92	
105:1.2DL+1.2FL1+1.2TL+1.6LL	709.31	-10.58	-160.58	101.04	3.35	52.79	157.18	51.60	150.48	44.90	130.18	104.39	97.69	153.83	48.25	
106:1.2DL+1.2FL2+1.2TL+1.6LL	786.36	-1.22	-176.73	112.02	0.39	58.10	170.50	54.31	169.73	53.53	143.50	112.41	111.63	170.12	53.92	
107:1.4DL+1.4FL1+1.4PL	780.55	-7.84	-173.77	111.19	2.48	57.12	170.80	56.55	165.83	51.58	143.80	113.68	108.71	168.32	54.07	
108:1.4DL+1.4FL2+1.4PL	870.44	3.09	-192.62	123.99	0.98	63.32	188.29	61.65	186.34	59.70	161.29	124.97	123.02	187.32	60.68	
109:1.2DL+1.2FL1+1.6LL+1.2PL	787.09	-6.23	-168.44	112.12	1.97	55.37	169.47	58.72	165.52	54.78	142.47	114.10	110.15	167.50	56.75	
110:1.2DL+1.2FL2+1.6LL+1.2PL	864.14	3.15	-184.59	123.10	1.00	60.68	184.77	63.41	182.78	61.42	157.77	124.09	122.10	183.78	62.42	
111:1.2DL+1.2FL1+1.2TL+1.6LL+1.2	789.46	-30.54	-215.08	112.46	9.67	70.70	192.83	51.42	173.49	32.09	165.83	122.13	102.79	183.16	41.76	
112:1.2DL+1.2FL2+1.2TL+1.6LL+1.2	866.51	-21.18	-231.25	123.43	6.71	76.02	206.16	54.12	192.75	40.71	179.16	130.14	116.73	199.46	47.42	
113:1.2DL+0.8WLX1	367.50	-7.47	-38.04	52.35	2.37	12.50	67.22	42.21	62.49	37.48	40.22	54.72	49.99	64.86	39.85	
114:1.2DL+0.8WLX2	366.17	16.72	-36.85	52.16	5.29	12.11	69.57	45.34	58.98	34.75	42.57	57.46	46.87	64.28	40.05	
115:1.2DL+0.8WLZ1	485.16	-11.84	-107.45	69.11	3.75	35.32	108.18	37.54	100.68	30.04	81.18	72.86	65.36	104.43	33.79	
116:1.2DL+0.8WLZ2	248.52	21.07	32.56	35.40	6.67	10.70	52.78	31.37	39.43	18.03	25.78	42.08	28.73	46.11	24.70	
117:1.2DL+1.6WLX1+0.5LL	395.51	-19.43	-44.73	56.34	6.15	14.71	77.19	47.78	64.90	35.49	50.19	62.49	50.20	71.05	41.64	
118:1.2DL+1.6WLX2+0.5LL	392.86	28.97	-42.35	55.96	9.17	13.92	79.05	51.21	60.71	32.87	52.05	65.13	46.79	69.88	42.04	
119:1.2DL+1.6WLZ1+0.5LL	630.83	-28.14	-183.54	89.86	8.91	60.33	159.10	38.43	141.29	20.62	132.10	98.77	80.96	150.20	29.53	
120:1.2DL+1.6WLZ2+0.5LL	157.54	37.68	96.46	22.44	11.93	31.71	66.08	2.66	42.22	0.00	39.08	34.37	21.11	54.15	1.33	
121:1.2DL+1.0SLX1+0.5LL+0.5FL1	493.98	-46.16	-67.58	70.37	14.61	22.22	107.19	62.76	77.97	33.54	80.19	84.98	55.76	92.58	48.15	
122:1.2DL+1.0SLX2+0.5LL+0.5FL1	504.88	62.88	-66.99	71.92	19.91	22.02	113.85	69.80	74.04	29.99	86.85	91.83	52.02	93.95	49.90	
123:1.2DL+1.0SLZ1+0.5LL+0.5FL1	744.95	-6.99	-199.60	106.12	2.21	65.61	173.95	42.72	169.52	38.29	146.95	108.34	103.91	171.74	40.51	
124:1.2DL+1.0SLZ2+0.5LL+0.5FL1	253.91	23.74	65.03	36.17	7.51	21.38	65.06	22.31	50.03	7.28	38.06	43.69	28.66	57.55	14.80	
125:0.9DL+1.6WLX1	329.89	-20.74	-29.27	46.99	6.56	9.62	63.18	43.94	50.05	30.81	36.18	53.56	40.43	56.62	37.38	
126:0.9DL+1.6WLX2	327.24	27.66	-26.90	46.62	8.76	8.84	64.21	46.53	46.70	29.02	37.21	55.37	37.86	55.46	37.78	
127:0.9DL+1.6WLZ1	565.21	-29.43	-168.09	80.51	9.32	55.26	145.09	34.57	126.45	15.94	118.09	89.83	71.20	135.77	25.26	
128:0.9DL+1.6WLZ2	91.92	36.37	111.93	13.09	11.51	36.79	61.40	0.00	38.37	0.00	34.40	30.70	19.19	49.89	0.00	
129:0.9DL+1.0SLX1	323.11	-51.06	-28.38	46.03	16.16	9.33	71.52	52.86	39.19	20.53	44.52	62.19	29.86	55.36	36.70	
130:0.9DL+1.0SLX2	334.02	57.98	-27.81	47.58	18.35	9.14	75.08	56.79	38.37	20.09	48.08	65.94	29.23	56.73	38.44	
131:0.9DL+1.0SLZ1	574.08	-11.90	-160.40	81.78	3.77	52.73	138.27	32.82	130.74	25.29	111.27	85.55	78.02	134.51	29.06	

MD-216-3310-EG-CV-CAL-1194				Mansourah Massarah Gold Project												
900707_GAU01_ZF01_CED01_33102 _OU100665189_R2				DESIGN CALCULATION ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION										Date: 09.11.2023		
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132:0.9DL+1.0SLZ2	83.05	18.83	104.23	11.83	5.96	34.26	52.06	0.00	40.13	0.00	25.06	26.03	20.07	46.10	0.00	

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Max :	870	63	112	**Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL cases) =**	**80.97 kN/m²**
Min :	83	-51	-231	**Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL cases) =**	**90.97 kN/m²**
				Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL+SL/WL cases) =	**148.54 kN/m²**
				Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL+WL/SL cases) =	**158.54 kN/m²**

Sample calculation of values of base pressure:

Load case : 202:1.0DL+1.0FL2

Maximum pressure, p_1 (pmax) =	87.75+5.97+30.30=	124.02 kN/m²	Maximum average Pressure along X, $p_{avx(max)}$ =	(124.02+63.42)/2= 93.72 kN/m²
p_2=	87.75+5.97-30.30=	63.42 kN/m²	Minimum average Pressure along X, $p_{avx(minx)}$ =	(51.49+112.08)/2= 81.78 kN/m²
p_3=	87.75-(5.97+30.30)=	51.49 kN/m²	Maximum average Pressure along Y, $p_{avy(max)}$ =	(124.02+112.08)/2= 118.05 kN/m²
p_4=	87.75-5.97+30.30=	112.08 kN/m²	Minimum average Pressure along Y, $p_{avy(minx)}$ =	(51.49+63.42)/2= 57.45 kN/m²
Maxm pressure excluding weight of soil (p_{max}-γh)=	124.02-1.5*18=	97.02 kN/m²		

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DESIGN OF FOUNDATION MKD. F2

Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m^2.

p_{2x} =pressure at the face of column at section 2x-2x in kN/m^2.

p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m^2.

BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.

BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.

SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.

SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((0.65+2*0.5*0.376)+(0.55+2*0.5*0.376))*2

= 3.904 m

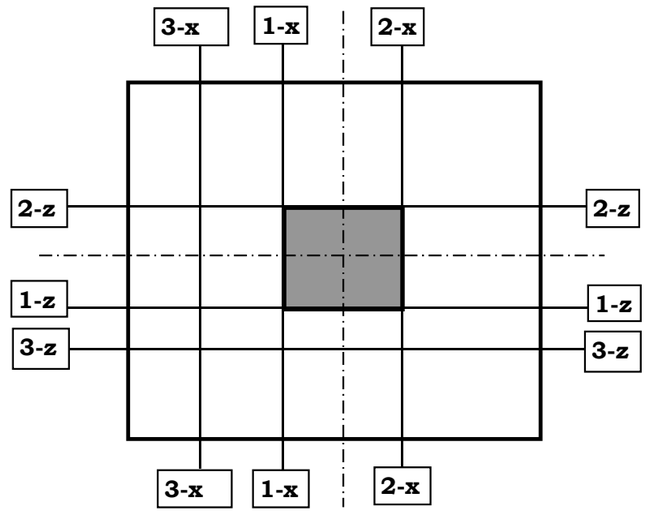
Punching Area= ((0.65+2*0.5*0.376)*(0.55+2*0.5*0.376))

= 0.95 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.45*24+1.05*18)

= 29.70 kN/m^2

Area of footing, A_f= 7.02 m^2



Load Case	Load Combination	p_{1x} kN/m2	p_{2x} kN/m2	p_{3x} kN/m2	$B.M_{1x}$ kNm	$B.M_{2x}$ kNm	$S.F_{3x}$ kN	$S.F_{(punching)}$ kN
Ultimate Conditions								
101	1.4DL+1.4FL1	99.05	96.69	100.47	37.73	33.89	45.52	413.79
102	1.4DL+1.4FL2	112.69	108.67	115.11	45.82	39.26	55.31	491.51
103	1.2DL+1.2FL1+1.6LL	101.75	99.66	103.01	39.00	35.59	47.05	430.99
104	1.2DL+1.2FL2+1.6LL	113.44	109.91	115.56	45.93	40.19	55.43	497.61
105	1.2DL+1.2FL1+1.2TL+1.6LL	101.85	100.23	102.82	38.79	36.16	46.78	433.04
106	1.2DL+1.2FL2+1.2TL+1.6LL	112.11	111.93	112.23	43.40	43.09	52.30	499.66
107	1.4DL+1.4FL1+1.4PL	111.79	110.60	112.51	43.79	41.84	52.79	494.64
108	1.4DL+1.4FL2+1.4PL	124.23	123.76	124.51	49.92	49.15	60.16	572.37
109	1.2DL+1.2FL1+1.6LL+1.2PL	112.60	111.65	113.17	44.07	42.52	53.13	500.30
110	1.2DL+1.2FL2+1.6LL+1.2PL	123.33	122.86	123.62	49.45	48.67	59.60	566.92
111	1.2DL+1.2FL1+1.2TL+1.6LL+1.	114.79	110.13	117.60	47.27	39.68	57.07	502.35
112	1.2DL+1.2FL2+1.2TL+1.6LL+1.	125.05	121.82	127.00	51.87	46.61	62.58	568.97

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113:1.2DL+0.8WLX1	52.92	51.79	53.61	12.83	10.97	15.49	137.49
114:1.2DL+0.8WLX2	53.44	50.89	54.98	13.88	9.72	16.79	136.34
115:1.2DL+0.8WLZ1	70.01	68.21	71.10	22.17	19.23	26.76	239.23
116:1.2DL+0.8WLZ2	37.01	33.80	38.95	5.62	0.38	6.85	34.61
117:1.2DL+1.6WLX1+0.5LL	57.82	54.87	59.61	16.41	11.59	19.84	161.71
118:1.2DL+1.6WLX2+0.5LL	58.17	53.75	60.83	17.39	10.20	21.07	159.42
119:1.2DL+1.6WLZ1+0.5LL	92.01	87.72	94.59	35.10	28.11	42.40	365.18
120:1.2DL+1.6WLZ2+0.5LL	29.34	26.14	31.26	1.57	0.00	1.97	0.00
121:1.2DL+1.0SLX1+0.5LL+0.5FL1	73.89	66.85	78.13	27.10	15.63	32.82	246.85
122:1.2DL+1.0SLX2+0.5LL+0.5FL1	76.72	67.13	82.50	29.99	14.37	36.37	256.28
123:1.2DL+1.0SLZ1+0.5LL+0.5FL1	106.66	105.59	107.30	41.02	39.28	49.45	463.86
124:1.2DL+1.0SLZ2+0.5LL+0.5FL1	37.98	34.37	40.17	6.35	0.45	7.74	39.27
125:0.9DL+1.6WLX1	48.58	45.41	50.48	11.66	6.51	14.13	104.97
126:0.9DL+1.6WLX2	48.72	44.51	51.26	12.32	5.45	14.95	102.68
127:0.9DL+1.6WLZ1	82.76	78.27	85.46	30.35	23.04	36.68	308.44
128:0.9DL+1.6WLZ2	26.33	23.56	28.00	0.00	0.00	0.00	0.00
129:0.9DL+1.0SLX1	49.92	42.13	54.61	14.92	2.23	18.17	99.11
130:0.9DL+1.0SLX2	52.00	43.17	57.33	16.60	2.19	20.22	108.54
131:0.9DL+1.0SLZ1	82.69	80.88	83.78	28.84	25.88	34.79	316.11
132:0.9DL+1.0SLZ2	23.77	22.33	24.63	0.00	0.00	0.00	0.00

Sample calculation of BM & SF

Load case : 101:1.4DL+1.4FL1

p_{1x} = 102.76-(102.76-92.98)*1.025/2.7 = 99.05 kN/m^2 Overhang = 1.025 m

p_{2x} = 92.98+(102.76-92.98)*1.025/2.7 = 96.69 kN/m^2 Effective depth = 0.392 m

p_{3x} = 102.76-(102.76-92.98)*(1.025-0.392)/2.7 = 100.47 kN/m^2

BM_{1x} = (99.05-29.7)*1.025*1.025/2+0.5*(102.76-99.05)*1.025*1.025*0.67 = 37.73 kN-m/m

BM_{2x} = (96.69-29.7)*1.025*1.025/2-0.5*(-92.98+96.69)*1.025*1.025*0.67 = 33.89 kN-m/m

SF_{3x} = (100.47-29.7)*(1.025-0.392)+0.5*(102.76-100.47)*(1.025-0.392) = 45.52 kN/m

$S.F_{(punching)}$ = (7.02-0.95)*(97.87-29.7) = 413.79 kN

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Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	51.87	0.34	0.45	0.001	423.30	705.60	16ϕT	284.95	300	250	804.25

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc)/bd = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 62.58 kN/m

Shear stress , τ_v= (62.58*1000)/(1000*392)

= 0.16 N/mm^2 **O.K.**

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 572.37 kN

Strength of Concrete in shear Vc1 = 4062.59 kN **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 3024.11 kN **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 3017.93 kN **Table 22.6.5.2_ (c)**

Design Vc= 3017.93 kN

Allowable shear stress, (ϕ Vc)/bd = 1.48 N/mm^2

Design punching shear stress, τ = (572.37*1000)/(3904*392)

= 0.37 N/mm^2 **No shear reinforcement reqd.**

Calculation of Bending Moment and Shear Force in Z direction :

p1z =pressure at the face of column at section 1z-1z in kN/m^2.
p2z =pressure at the face of column at section 2z-2z in kN/m^2.
p3z =pressure at a distance d from the face of column at section 3z-3z in kN/m^2.
B.M1z =Moment at the face of column at section 1z-1z in kN-m/m.
B.M2z =Moment at the face of column at section 2z-2z in kN-m/m.
S.F 3z = Oneway Shear Force at a distance d from the face of column at section 3z-3z in kN/m.
S.F (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Load Case	Load Combination	p_{1z} kN/m2	p_{2z} kN/m2	p_{3z} kN/m2	$B.M_{1z}$ kNm	$B.M_{2z}$ kNm	$S.F_{3z}$ kN
Ultimate Conditions							
101:1.4DL+1.4FL1		105.53	90.20	116.00	49.88	21.73	61.88
102:1.4DL+1.4FL2		119.65	92.18	131.92	59.02	17.03	73.21
103:1.2DL+1.2FL1+1.6LL		108.63	94.50	119.46	51.86	19.10	64.32
104:1.2DL+1.2FL2+1.6LL		120.73	102.37	133.09	59.68	15.73	74.03
105:1.2DL+1.2FL1+1.2TL+1.6LL		112.21	95.20	127.48	57.99	16.15	72.01
106:1.2DL+1.2FL2+1.2TL+1.6LL		124.31	90.84	141.11	65.82	14.50	81.72
107:1.4DL+1.4FL1+1.4PL		123.28	96.28	139.80	65.01	17.11	80.71
108:1.4DL+1.4FL2+1.4PL		137.39	78.09	155.71	74.15	16.45	92.04
109:1.2DL+1.2FL1+1.6LL+1.2PL		123.84	75.42	139.86	64.82	16.08	80.46
110:1.2DL+1.2FL2+1.6LL+1.2PL		135.94	98.65	153.49	72.65	17.35	90.17
111:1.2DL+1.2FL1+1.2TL+1.6LL+1.		127.42	49.12	147.86	70.95	12.14	88.14
112:1.2DL+1.2FL2+1.2TL+1.6LL+1.		139.52	75.41	161.51	78.79	17.65	97.86





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113:1.2DL+0.8WLX1	55.00	70.54	58.62	16.76	16.89	20.79
114:1.2DL+0.8WLX2	54.73	90.45	58.23	16.51	17.59	20.48
115:1.2DL+0.8WLZ1	76.58	47.36	86.80	34.43	15.54	42.78
116:1.2DL+0.8WLZ2	37.67	55.97	40.77	7.16	12.45	8.92
117:1.2DL+1.6WLX1+0.5LL	59.46	66.60	63.71	19.71	14.72	24.45
118:1.2DL+1.6WLX2+0.5LL	58.90	88.31	62.93	19.20	16.17	23.82
119:1.2DL+1.6WLZ1+0.5LL	102.63	41.03	120.08	55.05	8.95	68.43
120:1.2DL+1.6WLZ2+0.5LL	33.33	47.16	40.97	9.23	12.58	11.59
121:1.2DL+1.0SLX1+0.5LL+0.5FL1	75.06	75.90	81.49	30.00	17.99	37.21
122:1.2DL+1.0SLX2+0.5LL+0.5FL1	76.58	98.42	82.95	30.74	19.58	38.13
123:1.2DL+1.0SLZ1+0.5LL+0.5FL1	120.01	49.54	138.98	65.65	15.20	81.55
124:1.2DL+1.0SLZ2+0.5LL+0.5FL1	40.70	53.75	46.88	11.71	13.24	14.61
125:0.9DL+1.6WLX1	49.04	67.22	51.82	12.83	15.64	15.91
126:0.9DL+1.6WLX2	48.49	82.30	51.05	12.32	16.20	15.29
127:0.9DL+1.6WLZ1	92.20	45.24	108.19	48.17	13.83	59.89
128:0.9DL+1.6WLZ2	30.22	50.13	37.44	7.20	14.28	9.06
129:0.9DL+1.0SLX1	48.00	72.15	50.70	12.20	18.16	15.14
130:0.9DL+1.0SLX2	49.52	88.05	52.16	12.95	18.85	16.06
131:0.9DL+1.0SLZ1	92.94	52.90	108.19	47.85	16.82	59.48
132:0.9DL+1.0SLZ2	27.93	44.54	34.59	5.46	11.52	6.91

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Sample calculation of BM & SF

Load case : 101:1.4DL+1.4FL1

p_{1z} = 134.08-(134.08-61.65)*1.025/2.6= 105.53 kN/m^2

p_{2z} = 61.65+(134.08-61.65)*1.025/2.6= 90.20 kN/m^2

p_{3z} = 134.08-(134.08-61.65)*(1.025-0.376)/2.6= 116.00 kN/m^2

Overhang = 1.025 m

Effective depth = 0.376 m

BM_{1z} = (105.53-29.7)*1.025*1.025/2+0.5*(134.08-105.53)*1.025*1.025*0.67 = 49.88 kN-m/m

BM_{2z} = (90.20-29.7)*1.025*1.025/2-0.5*(-61.65+90.20)*1.025*1.025*0.67 = 21.73 kN-m/m

SF_{3z} = (116.00-29.7)*(1.025-0.376)+0.5*(134.08-116.00)*(1.025-0.376) = 61.88 kN/m

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	78.79	0.557	0.743	0.0018	702.39	676.80	16ϕT	286.25	300	250	804.2

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc) = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 97.86 kN/m

Shear stress , τ_v= (97.86*1000)/(1000*376)

= 0.26 N/mm^2 **O.K.**

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :

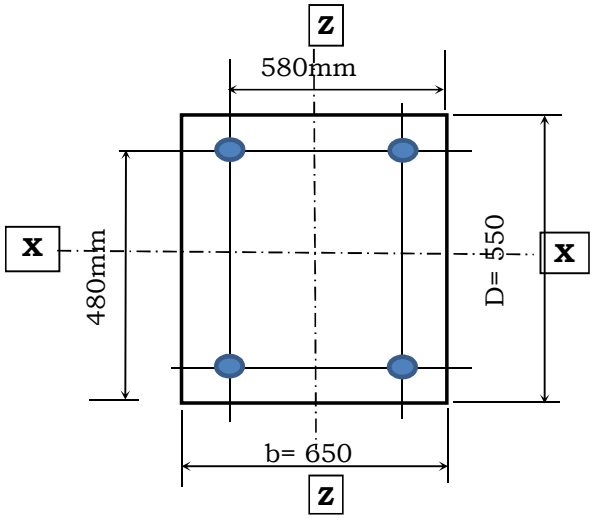
At Bottom **16ϕT** **250** c/c

Along Z Direction :

At Bottom **16ϕT** **250** c/c

MA'ADEN معادن		Outotec / LARSEN & TOUBRO
MD-216-3310-EG-CV-CAL-1194	Mansourah Massarah Gold Project	
900707_GAU01_ZF01_CED01_33102 _OU100665189_R2	DESIGN CALCULATION 3310 ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION	Date: 09.11.2023
O19045_450_C_K_ 24085 _001_R2	AS BUILT	Rev: D01

DESIGN OF COLUMN MKD C2 (STAAD Node -7093)



Input:

Clear cover to reinf.=	50mm	Table 20.6.1.3.1
Dia. of main reinf.=	20mm	
Dia. of link=	10mm	
Effective cover =	70mm	
Grade of Concrete (f_c') =	M35	
Grade of Steel (f_y) =	Fy420	
Grade of Reinf(links).=	Fy420	
Min. Steel Ratio	0.50%	
Max. Steel Ratio	4.00%	
Self weight of Column =	1.201kN	

For Compression-controlled sections

Strength reduction factor ϕ= 0.65 Table 21.2.2

Unsupproted length of column about X (l_{ux}) =	1400mm
Unsupproted length of column about Z (l_{uz}) =	1400mm

For Columns not braced against sidesway, Where, Effective length factor K= 1.2

Slenderness ratio,	K l_{ux} / r_x =	10.58	< 22 SECTION 6.2.5a	Radius of gyration r_x=	159	mm
	K l_{uz} / r_z =	8.96	< 22 SECTION 6.2.5a	Radius of gyration r_z=	188	mm

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900707_GAU01_ZF01_CED01_33102 _OU100665189_R2	DESIGN CALCULATION 3310 ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION	Date: 09.11.2023
O19045_450_C_K_ 24085 _001_R2	AS BUILT	Rev: D01

γ_y =	0.78
γ_z =	0.82

Load Case	Load Combination	Factored Load From STAAD			Load at Column Base			Nominal Load/Moment			Kn = Pn/f_cbD (N/mm^2)
		Pu (T)	Hux (T)	Huz (T)	Pu (T)	Muz (Tm)	Mux (Tm)	Pn (T)	Mnz (Tm)	Mnx (Tm)	
101	1.4DL+1.4FL1	47.33	0.8	-6.0	47.45	1.2	8.3	73.0	1.8	12.8	0.06
102	1.4DL+1.4FL2	56.32	1.4	-7.0	56.44	2.0	9.8	86.8	3.1	15.0	0.07
103	1.2DL+1.2FL1+1.6LL	49.32	0.7	-6.2	49.44	1.0	8.6	76.1	1.6	13.3	0.06
104	1.2DL+1.2FL2+1.6LL	57.02	1.2	-7.0	57.14	1.7	9.8	87.9	2.7	15.1	0.07
105	1.2DL+1.2FL1+1.2TL+1.6LL	49.56	-0.6	-8.7	49.68	0.8	12.2	76.4	1.2	18.7	0.06
106	1.2DL+1.2FL2+1.2TL+1.6LL	57.26	-0.1	-9.6	57.38	0.1	13.4	88.3	0.1	20.6	0.07
107	1.4DL+1.4FL1+1.4PL	56.68	-0.4	-9.4	56.80	0.6	13.2	87.4	0.9	20.2	0.07
108	1.4DL+1.4FL2+1.4PL	65.67	0.2	-10.4	65.79	0.2	14.6	101.2	0.4	22.4	0.08
109	1.2DL+1.2FL1+1.6LL+1.2PL	57.33	-0.3	-9.1	57.45	0.5	12.7	88.4	0.7	19.6	0.07
110	1.2DL+1.2FL2+1.6LL+1.2PL	65.04	0.2	-10.0	65.16	0.2	14.0	100.2	0.4	21.5	0.08
111	1.2DL+1.2FL1+1.2TL+1.6LL+1.2PL	57.57	-1.7	-11.6	57.69	2.3	16.3	88.8	3.6	25.0	0.07
112	1.2DL+1.2FL2+1.2TL+1.6LL+1.2PL	65.28	-1.1	-12.5	65.40	1.6	17.5	100.6	2.5	26.9	0.08
113	1.2DL+0.8WLX1	15.38	-0.4	-2.1	15.50	0.6	2.9	23.8	0.9	4.4	0.02
114	1.2DL+0.8WLX2	15.24	0.9	-2.0	15.36	1.3	2.8	23.6	1.9	4.3	0.02
115	1.2DL+0.8WLZ1	27.14	-0.6	-5.8	27.26	0.9	8.1	41.9	1.4	12.5	0.03
116	1.2DL+0.8WLZ2	3.48	1.1	1.8	3.60	1.6	2.5	5.5	2.5	3.8	0.00
117	1.2DL+1.6WLX1+0.5LL	18.18	-1.1	-2.4	18.30	1.5	3.4	28.1	2.3	5.2	0.02
118	1.2DL+1.6WLX2+0.5LL	17.91	1.6	-2.3	18.03	2.2	3.2	27.7	3.4	4.9	0.02
119	1.2DL+1.6WLZ1+0.5LL	41.71	-1.5	-9.9	41.83	2.1	13.9	64.4	3.3	21.4	0.05
120	1.2DL+1.6WLZ2+0.5LL	5.62	2.0	5.2	5.74	2.9	7.3	8.8	4.4	11.2	0.01
121	1.2DL+1.0SLX1+0.5LL+0.5FL1	28.02	-2.5	-3.7	28.14	3.5	5.1	43.3	5.4	7.9	0.03

MD-216-3310-EG-CV-CAL-1194		Mansourah Massarah Gold Project								
900707_GAU01_ZF01_CED01_33102 _OU100665189_R2		DESIGN CALCULATION 3310 ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION					Date: 09.11.2023			
O19045_450_C_K_ 24085 _001_R2		AS BUILT					Rev: D01			
122:1.2DL+1.0SLX2+0.5LL+0.5FL1	29.11	3.4	-3.6	29.23	4.8	5.1	45.0	7.3	7.8	0.04
123:1.2DL+1.0SLZ1+0.5LL+0.5FL1	53.12	-0.4	-10.8	53.24	0.5	15.1	81.9	0.8	23.2	0.07
124:1.2DL+1.0SLZ2+0.5LL+0.5FL1	4.02	1.3	3.5	4.14	1.8	4.9	6.4	2.8	7.6	0.01
125:0.9DL+1.6WLX1	11.61	-1.1	-1.6	11.73	1.6	2.2	18.1	2.4	3.4	0.01
126:0.9DL+1.6WLX2	11.35	1.5	-1.5	11.47	2.1	2.0	17.6	3.2	3.1	0.01
127:0.9DL+1.6WLZ1	35.15	-1.6	-9.1	35.27	2.2	12.7	54.3	3.4	19.6	0.04
128:0.9DL+1.6WLZ2	12.18	2.0	6.1	12.30	2.8	8.5	18.9	4.2	13.0	0.02
129:0.9DL+1.0SLX1	10.94	-2.8	-1.5	11.06	3.9	2.1	17.0	5.9	3.3	0.01
130:0.9DL+1.0SLX2	12.03	3.1	-1.5	12.15	4.4	2.1	18.7	6.8	3.2	0.01
131:0.9DL+1.0SLZ1	36.03	-0.6	-8.7	36.15	0.9	12.1	55.6	1.4	18.7	0.04
132:0.9DL+1.0SLZ2	13.07	1.0	5.6	13.19	1.4	7.9	20.3	2.2	12.1	0.02

Reinforcement: Provide,

4-Nos	20 mm Dia
4-Nos	20 mm Dia
4-Nos	20 mm Dia

1257.14 Sq mm (At Corner)
1257.14 Sq mm (Along Z axis)
1257.14 Sq mm (Along Y axis)

A_{sc} pvd. = 3771.43 Sq mm

Total A_{sc} = 1.1 %

$\beta = P_n / P_0$ (range is 0.55 to 0.70)

Where, $P_0 = 0.85 f'_c (A_g - A_{sc}) + f_y A_{sc}$ = 1210.743 T SECTION 22.4.2.2

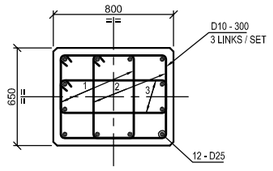
Load Case	From Chart of SP17(M)-R35-420.8		M_{noz}	M_{nox}	β	$\alpha_n = \log(0.5)/\log(\beta)$	Interaction ratio	Remarks
	$R_{nz}=M_{noz}/f'_c b^2 D$	$R_{nx}=M_{nox}/f'_c b^2 D$						
101	0.05	0.05	40.7	34.4	0.55	1.16	0.35	O.K
102	0.05	0.05	40.7	34.4	0.55	1.16	0.43	O.K
103	0.05	0.05	40.7	34.4	0.55	1.16	0.35	O.K
104	0.05	0.05	40.7	34.4	0.55	1.16	0.43	O.K
105	0.05	0.05	40.7	34.4	0.55	1.16	0.51	O.K
106	0.05	0.05	40.7	34.4	0.55	1.16	0.55	O.K
107	0.05	0.05	40.7	34.4	0.55	1.16	0.55	O.K
108	0.05	0.05	40.7	34.4	0.55	1.16	0.61	O.K

MD-216-3310-EG-CV-CAL-1194			Mansourah Massarah Gold Project					
900707_GAU01_ZF01_CED01_33102 _OU100665189_R2			DESIGN CALCULATION 3310 ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION				Date: 09.11.2023	
O19045_450_C_K_ 24085 _001_R2			AS BUILT				Rev: D01	
109	0.05	0.05	40.7	34.4	0.55	1.16	0.53	O.K
110	0.05	0.05	40.7	34.4	0.55	1.16	0.58	O.K
111	0.05	0.05	40.7	34.4	0.55	1.16	0.75	O.K
112	0.05	0.05	40.7	34.4	0.55	1.16	0.79	O.K
113	0.05	0.05	40.7	34.4	0.55	1.16	0.10	O.K
114	0.05	0.05	40.7	34.4	0.55	1.16	0.12	O.K
115	0.05	0.05	40.7	34.4	0.55	1.16	0.33	O.K
116	0.05	0.05	40.7	34.4	0.55	1.16	0.12	O.K
117	0.05	0.05	40.7	34.4	0.55	1.16	0.15	O.K
118	0.05	0.05	40.7	34.4	0.55	1.16	0.16	O.K
119	0.05	0.05	40.7	34.4	0.55	1.16	0.63	O.K
120	0.05	0.05	40.7	34.4	0.55	1.16	0.35	O.K
121	0.05	0.05	40.7	34.4	0.55	1.16	0.28	O.K
122	0.05	0.05	40.7	34.4	0.55	1.16	0.32	O.K
123	0.05	0.05	40.7	34.4	0.55	1.16	0.65	O.K
124	0.05	0.05	40.7	34.4	0.55	1.16	0.22	O.K
125	0.05	0.05	40.7	34.4	0.55	1.16	0.11	O.K
126	0.05	0.05	40.7	34.4	0.55	1.16	0.11	O.K
127	0.05	0.05	40.7	34.4	0.55	1.16	0.58	O.K
128	0.05	0.05	40.7	34.4	0.55	1.16	0.40	O.K
129	0.05	0.05	40.7	34.4	0.55	1.16	0.17	O.K
130	0.05	0.05	40.7	34.4	0.55	1.16	0.19	O.K
131	0.05	0.05	40.7	34.4	0.55	1.16	0.51	O.K
132	0.05	0.05	40.7	34.4	0.55	1.16	0.33	O.K

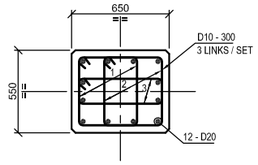
Links, 10mm dia @ 300 C/C

ATTACHMENT 1: Drawings

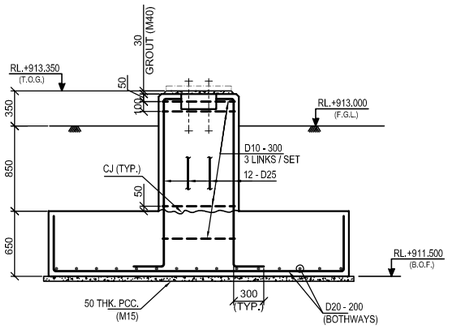
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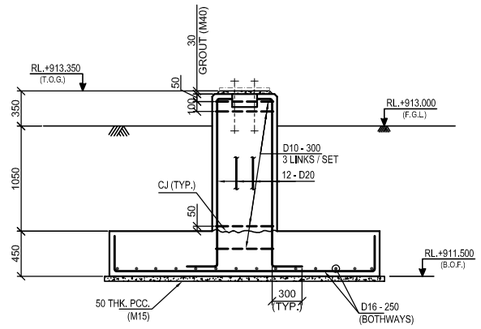
RC. DET. OF C1
SCALE 1:20



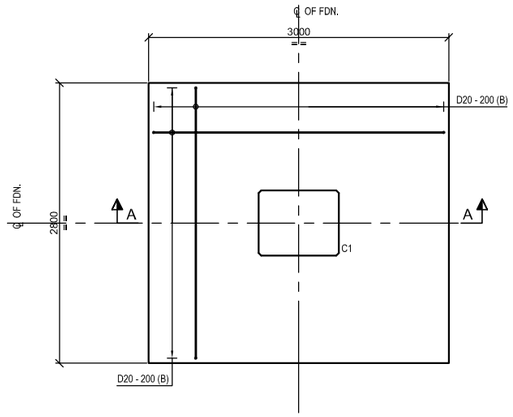
RC. DET. OF C2
SCALE 1:20



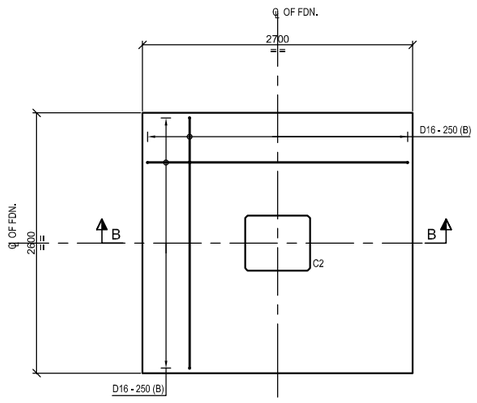
SECTION : A-A
SCALE 1:30



SECTION : B-B
SCALE 1:30



RC. DET. OF FOUNDATION
MKD. F1
SCALE 1:30



RC. DET. OF FOUNDATION
MKD. F2
SCALE 1:30

LEGEND :-

T.O.G - TOP OF GROUT
CJ - CONSTRUCTION JOINT
F.G.L - FINISHED GROUND LEVEL
P.C.C - PLAIN CEMENT CONCRETE
FDN. - FOUNDATION
B.O.F - BOTTOM OF FOUNDATION

NOTES:-

1. GRADE OF CONCRETE SHALL BE 35 MPa FOR RCC, 15 MPa FOR BLINDING CONCRETE.
2. GRADE OF REINFORCEMENT SHALL BE DEFORMED BAR 420 MPa (GRADE 60) - ASTM A615/A615M.
3. CLEAR COVER TO MAIN REINFORCEMENT SHALL BE COLUMN - 50MM,. FOUNDATION - 50 MM.
4. FOR ALL OTHER NOTES, REFER DRAWING NO, MD-216-3310-EG-CV-GAD-1173, MD-216-3310-EG-CV-GAD-1174.

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3310-EG-CV-GAD-1174	ROUGHER AND SCAVENGER FLOTATION								
MD-216-3310-EG-CV-GAD-1173	ROUGHER AND SCAVENGER FLOTATION								
MD-216-3310-EG-CV-GAD-1001	FLOTATION ROUGHER/SCAVENGER/CLEANER								
MD-216-3310-EG-ST-GAD-2163	ROUGHER AND SCAVENGER FLOTATION	A01	12/12/2019	ISSUED FOR REFERENCE	ABS	SJM	AVG	AG	

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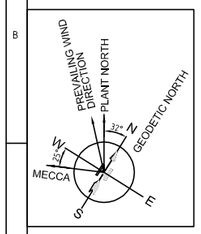
LARSEN & TOUBRO DRAWING NUMBER: 019045_450_C_D_24084_001 — REVISION: 0
OUTOTEC DRAWING NUMBER: 900707__GAU01__ZF01__CTB13__33101 — DOCUMENT ID: OU100665541 — REVISION: 0

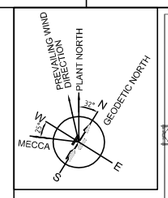
DRAWN	ABS	10/12/2019
CHECKED	SJM	11/12/2019
ENGINEER	AVG	11/12/2019
APPROVED	AG	12/12/2019
EMPLOYER		
SCALE	1:30, 1:20	A1

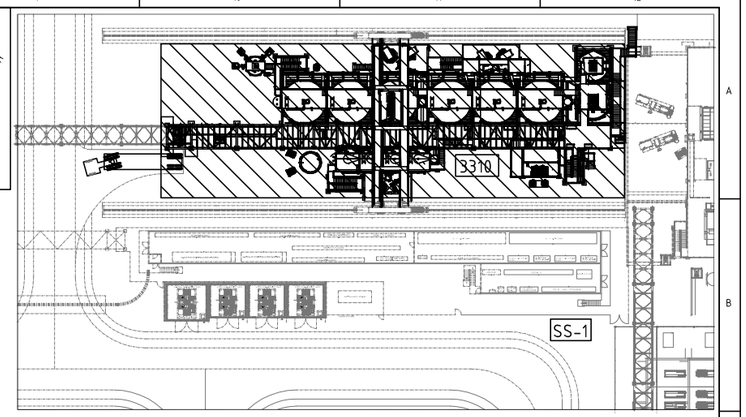
MANSOURAH - MASSARAH GOLD PROJECT
DETAIL DRAWING
ROUGHER AND SCAVENGER FLOTATION
RC DETAILS OF FOUNDATION

DRAWING NUMBER: MD-216-3310-EG-CV-DED-1184 — REV: A01

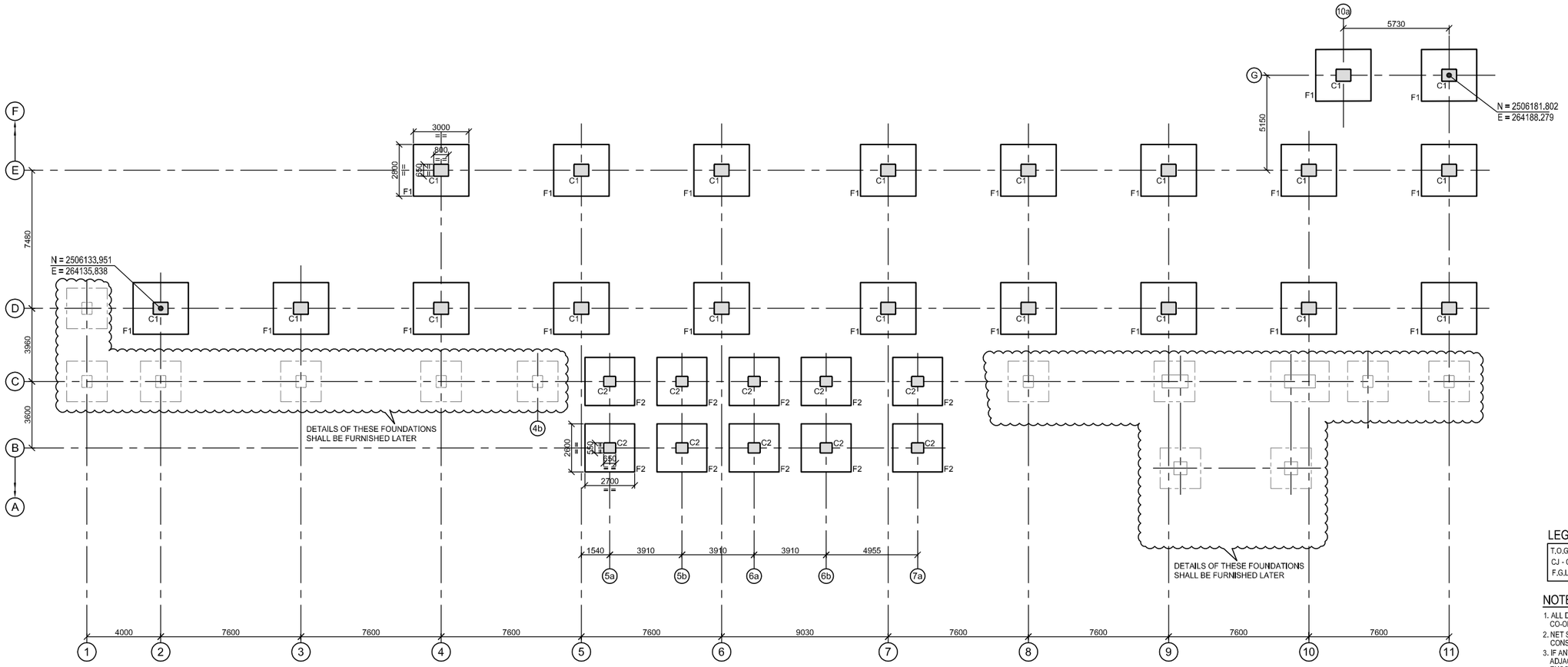
DRAWING NUMBER MD-216-3310-EG-CV-GAD-1173 REV A01







KEY PLAN
SCALE 1:500



LAYOUT OF FOUNDATION
SCALE 1:125

LEGEND :-

T.O.G - TOP OF GROUT
CJ - CONSTRUCTION JOINT
F.G.L - FINISHED GROUND LEVEL

NOTES:-

1. ALL DIMENSIONS ARE IN MILLIMETRES AND LEVELS AND CO-ORDINATES ARE IN METRES UNLESS OTHERWISE STATED.
2. NET SAFE BEARING CAPACITY OF SOIL AT BOTTOM OF FOUNDATION CONSIDER 450KN/ SQ. M
3. IF ANY PART OF FOUNDATION COMES ON BACKFILL ZONE OF ADJACENT FOUNDATION OR ANY LOOSE SOIL ARE ENCOUNTERED BELOW FOUNDATION LEVEL THE SAME SHALL BE REMOVED AND FILLED WITH PCC (M15).
4. THE CONTENT OF THIS DRAWING SHALL CONFORM TO THE PROVISIONS OF CIVIL, STRUCTURAL AND ARCHITECTURAL DESIGN CRITERIA DOC. NO. MD-216-1000-EG-ST-CRT-1001.

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3310-EG-CV-GAD-1174	ROUGHER AND SCAVENGER FLOTATION								
MD-216-3310-EG-CV-GAD-1001	FLOTATION ROUGHER/SCAVENGER/CLEANER								
MD-216-3310-EG-ST-GAD-2163	ROUGHER AND SCAVENGER FLOTATION	A01	10/12/2019	ISSUED FOR REFERENCE	ABS	SJM	AVG	AG	

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Saudi Arabian Mining Company شركة التعدين العربية السعودية

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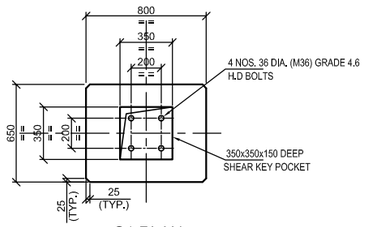
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OUTOTEC DRAWING NUMBER: 900707__GAU01__ZF01__MTL01__33128 — DOCUMENT ID: OU100665308 — REVISION: 0

DRAWN	ABS	14/11/2019
CHECKED	SJM	09/12/2019
ENGINEER	AVG	25/11/2019
APPROVED	AG	10/12/2019
EMPLOYER		
SCALE	1:125	A1

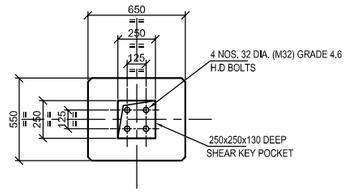
MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
ROUGHER AND SCAVENGER FLOTATION
LAYOUT & DETAIL OF FOUNDATION

DRAWING NUMBER: MD-216-3310-EG-CV-GAD-1173 — REV: A01

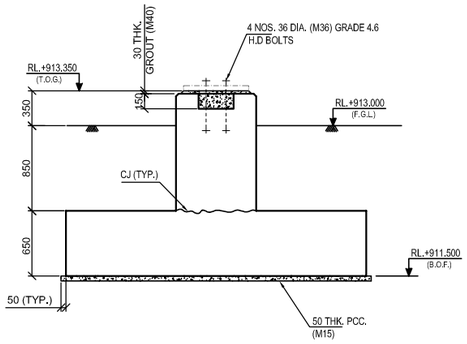
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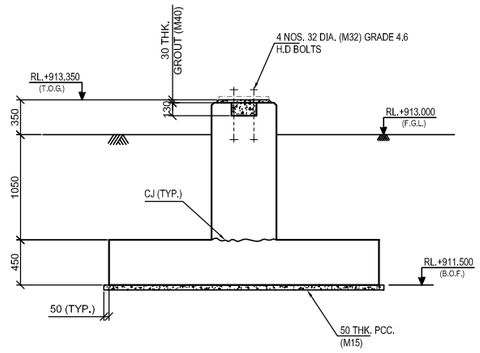
C1 PLAN
SCALE 1:20



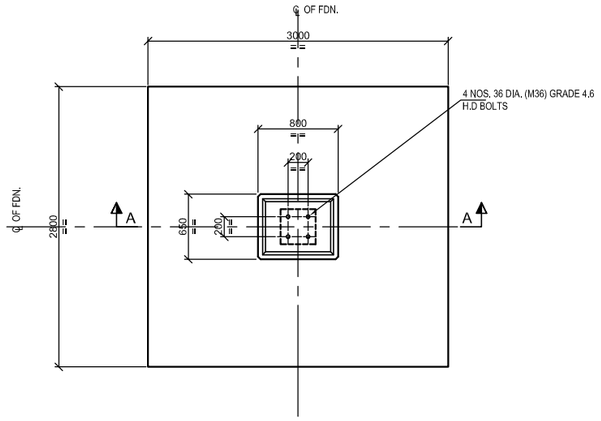
C2 PLAN
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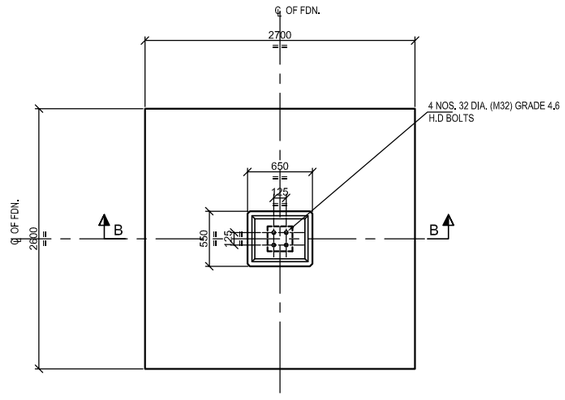
SECTION : A-A
SCALE 1:30



SECTION : B-B
SCALE 1:30



PLAN OF FOUNDATION
MKD. F1
(20 NOS. REQD.)
SCALE 1:30



PLAN OF FOUNDATION
MKD. F2
(10 NOS. REQD.)
SCALE 1:30

LEGEND :-

T.O.G - TOP OF GROUT
CJ - CONSTRUCTION JOINT
F.G.L - FINISHED GROUND LEVEL
P.C.C - PLAIN CEMENT CONCRETE
FDN. - FOUNDATION
B.O.F - BOTTOM OF FOUNDATION

NOTES:-

1. FOR DETAIL OF ANCHOR BOLTS, REFER DRAWING NO. MD-216-1000-EG-CV-DED-1003.
2. FOR ALL OTHER NOTES, REFER DRAWING NO. MD-216-3310-EG-CV-GAD-1173.

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3310-EG-CV-GAD-1173	ROUGHER AND SCAVENGER FLOTATION								
MD-216-3310-EG-CV-GAD-1001	FLOTATION ROUGHER/SCAVENGER/CLEANER								
MD-216-3310-EG-ST-GAD-2163	ROUGHER AND SCAVENGER FLOTATION	A01	10/12/2019	ISSUED FOR REFERENCE	ABS	SJM	AVG	AG	

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LARSEN & TOUBRO DRAWING NUMBER: 019045_450_C_G_24083_002 — REVISION: 0
OUTOTEC DRAWING NUMBER: 900707__GAU01__ZF01__MTL01__33129 — DOCUMENT ID: OU100665309 — REVISION: 0

DRAWN	ABS	14/11/2019
CHECKED	SJM	09/12/2019
ENGINEER	AVG	25/11/2019
APPROVED	AG	10/12/2019
EMPLOYER		
SCALE	1:30, 1:20	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
ROUGHER AND SCAVENGER FLOTATION
LAYOUT & DETAIL OF FOUNDATION

DRAWING NUMBER: MD-216-3310-EG-CV-GAD-1174 — REV: A01